

BB

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RF

102.RF_Transceiver_SDR435_WMSS SH
103.RF_Transceiver_SDR435_TRX SH
107.LDO/APT_Module SH
108.MMPA_Phase5N#0
109.Phase5N#0_DUP
110.MMPA_Phase5N#1
111.PRX_LNA#0
112.TXM_NZ5596M
113.N77_N78_PAM
114.DRX_MHB_MIMO
115.DRX_LMHB
116.ASDIV_FBRX
135.WWAN_ANT_PART1
136.WWAN_ANT_PART2
144.Connectivity_ANT
145.Connectivity_WCN3950
150.GPS
152.NFC_SN100T

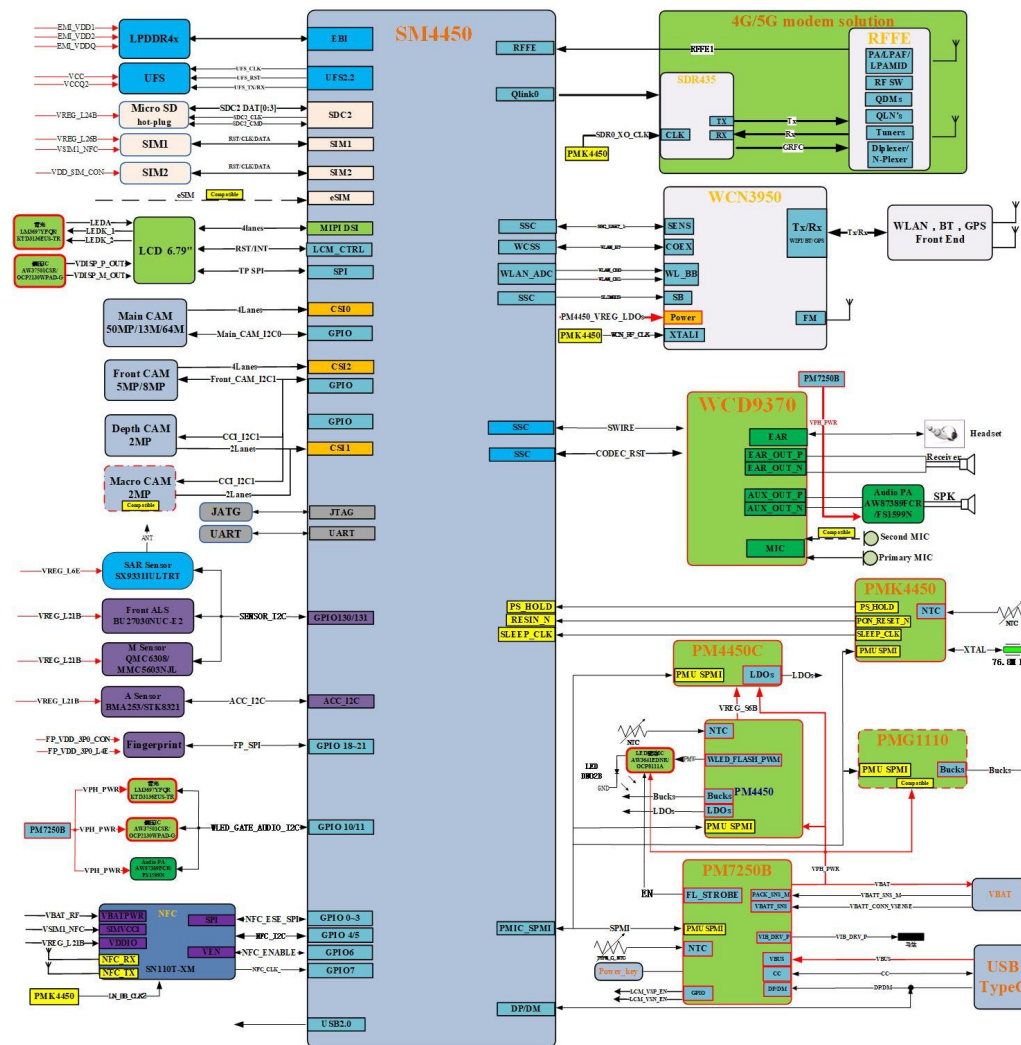
001_Revision history

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DOCUMENT NO.: 86029_1_12M14_20230417_1755		Size: D
DEPARTMENT: BB		DESIGNER: ZHAOYUKUN
		
Date: Monday, April 17, 2023	Sheet: 2 of 59	

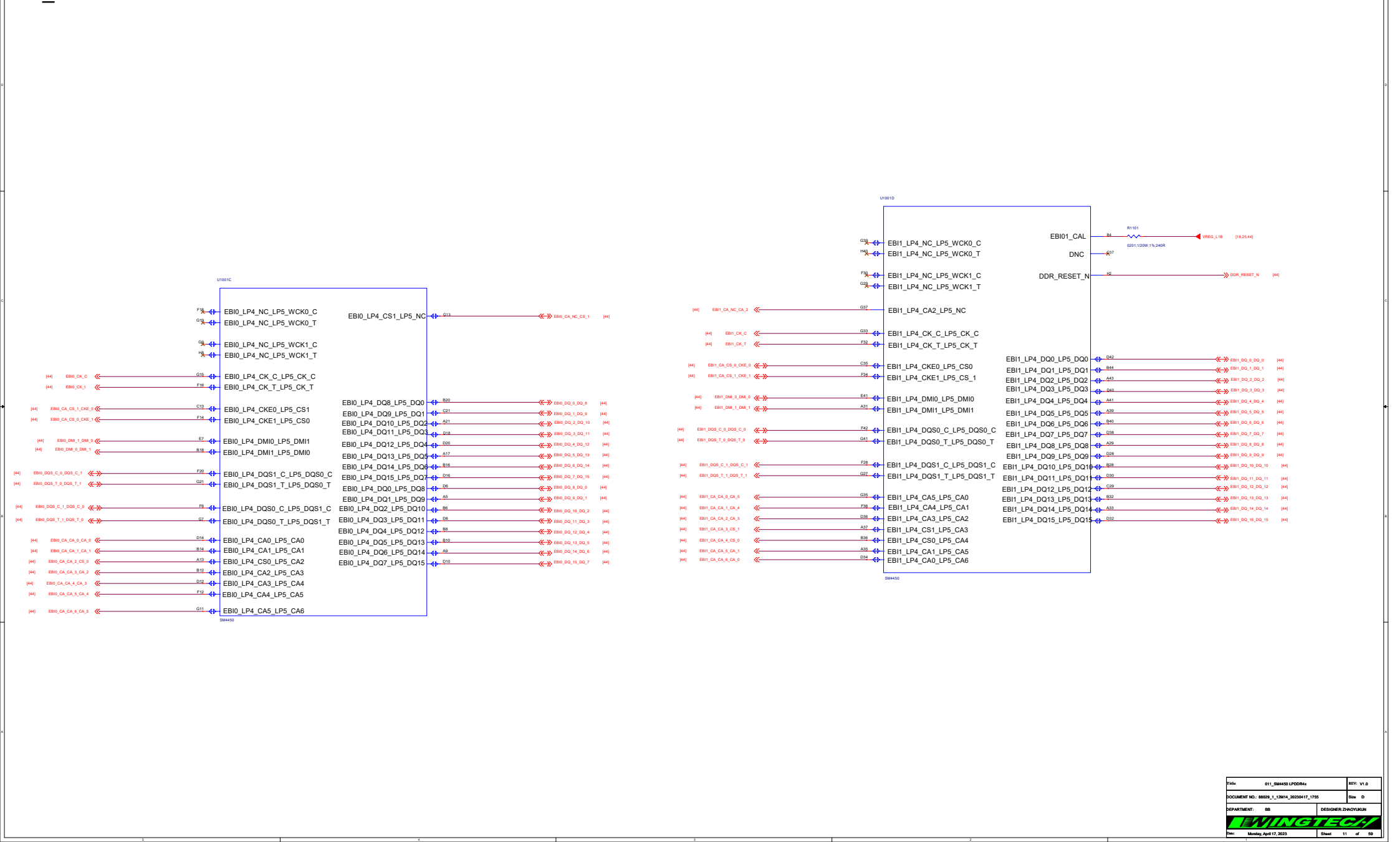
002_CHANGE LIST

Title: 002_CHANGE LIST		REV: V1.0
DOCUMENT NO.: 86029_1_12M14_20230417_1755		Size: D
DEPARTMENT: BB		DESIGNER: ZHAOYUKUN
		
Date: Monday, April 17, 2023	Sheet 3 of 58	

004_Block_Diagram



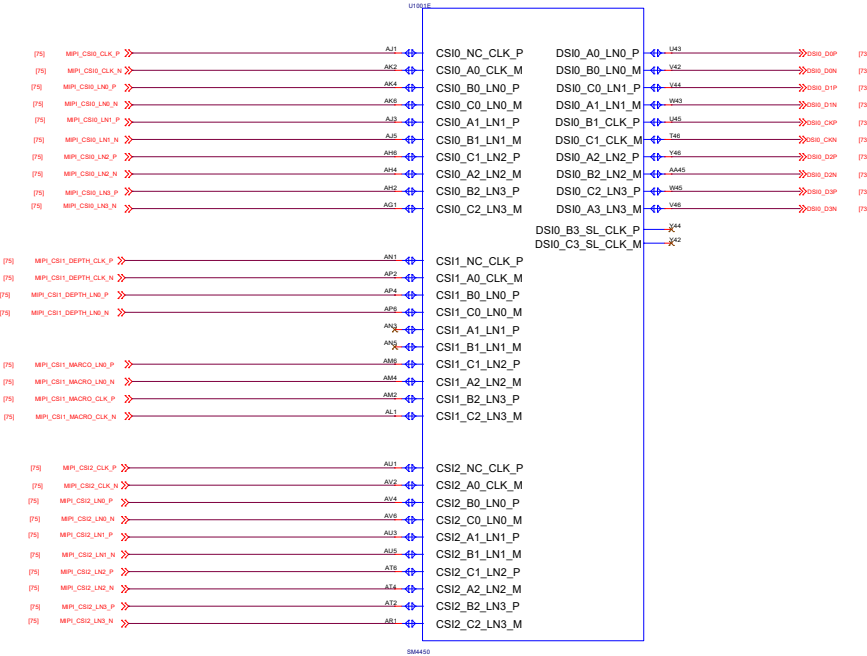
011_SM4450 LPDDR4x



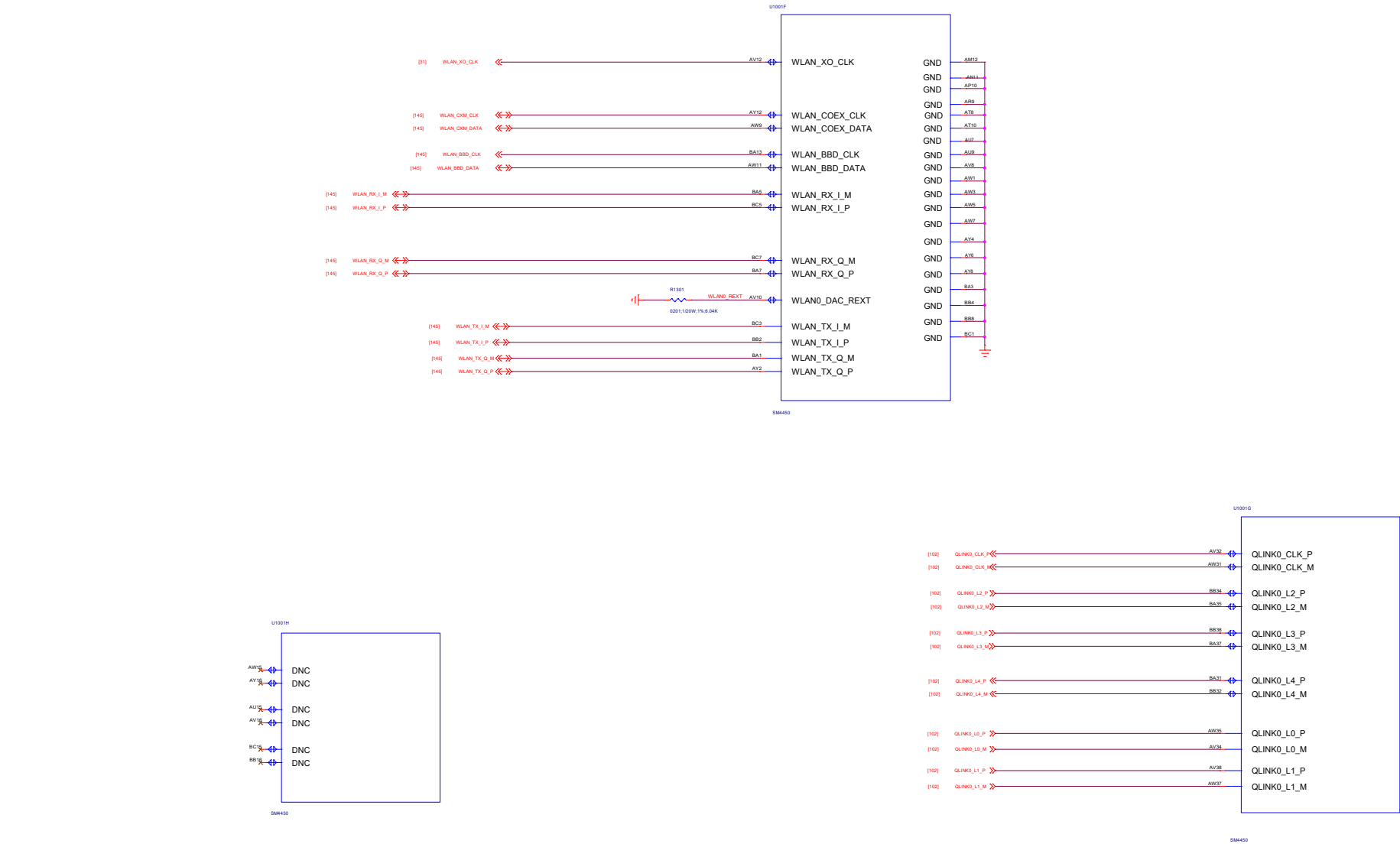
CSI0
M50M or
M64M or
M13M

CSI1
Depth2M
+Macro2M

CSI2
F5M or
F8M

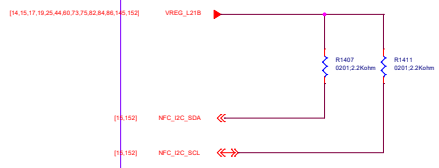


013_SM4450_IQ_QLINK

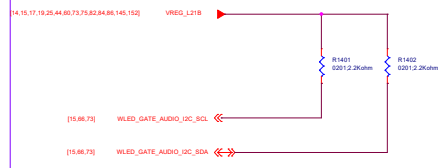


I2C BUS PULL-UPS

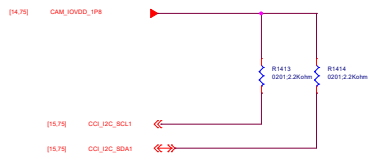
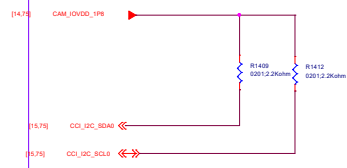
NFC



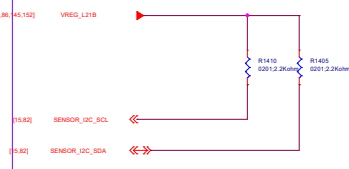
WLED&GATE



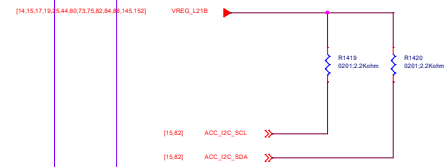
Camera



SENSOR

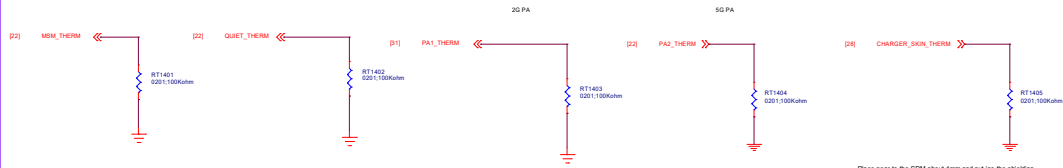


ACC I2C



NTC

SM4450-Thermistors

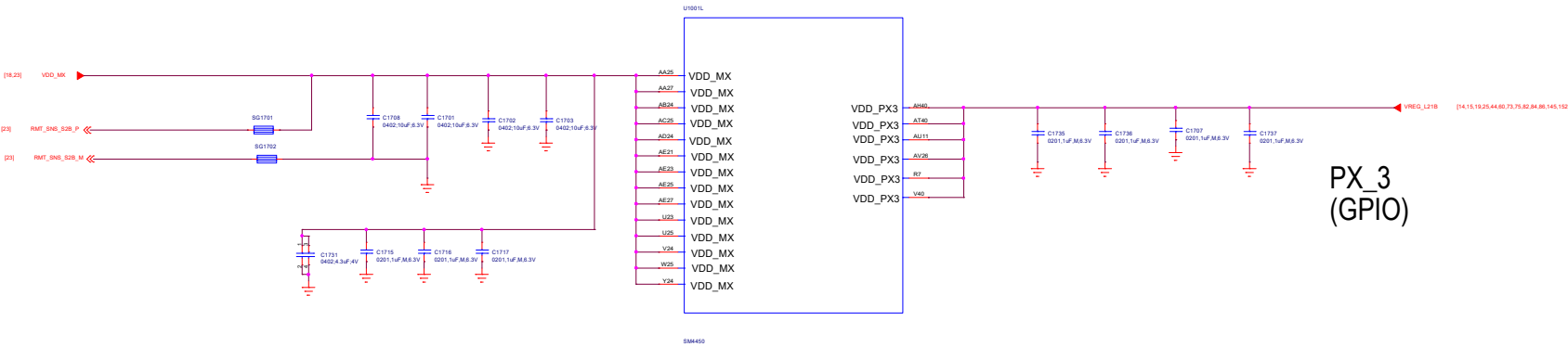


Place near to the SOM about 1mm and put into the shielding

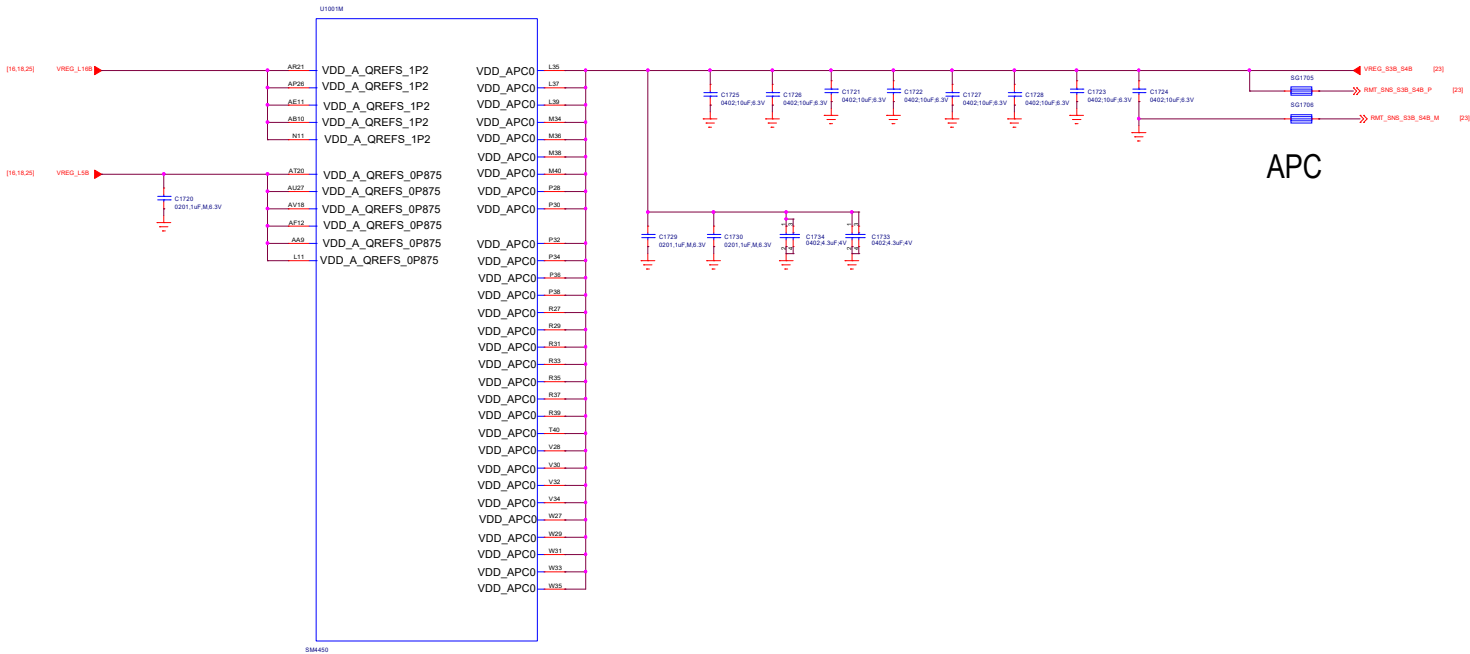
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DOCUMENT NO: 88029_1_12M14_20230417_1755	Size D	
DEPARTMENT: BB	DESIGNER: ZHAOYUKUN	
WINGTECH		
Date: Monday, April 17, 2023	Sheet 16 of 59	

017_SM4450_PWR_1

MX



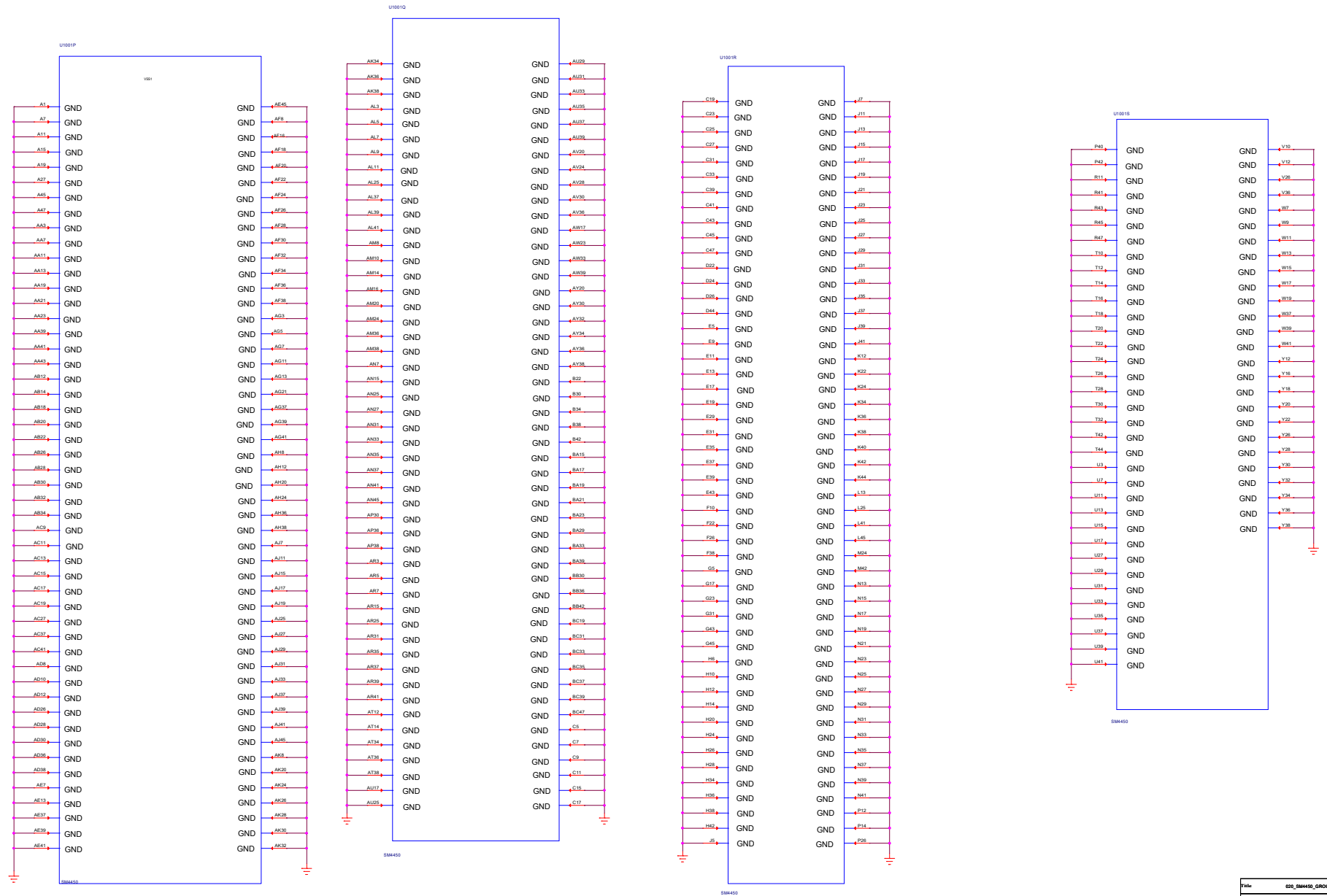
PX_3
(GPIO)

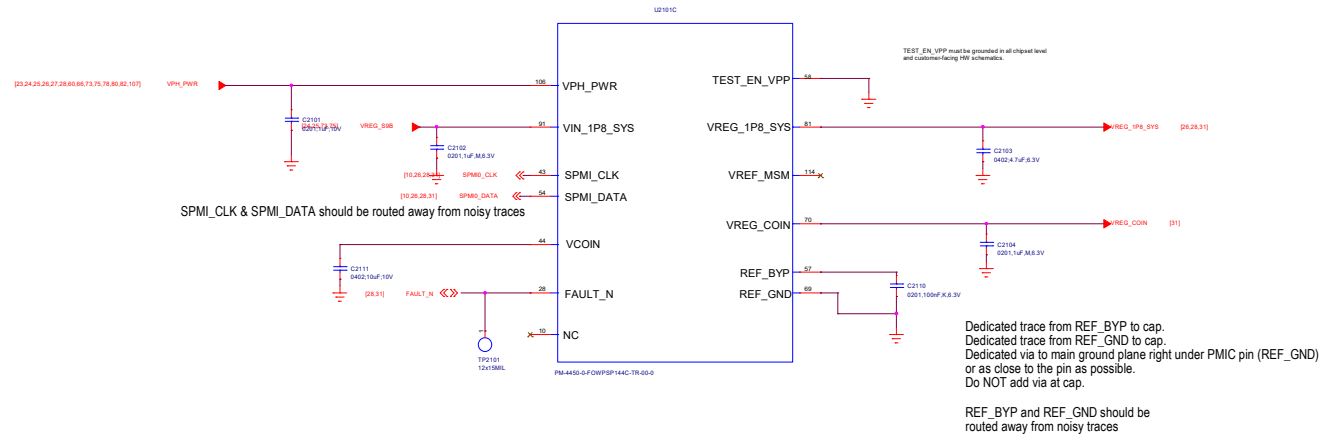


APC

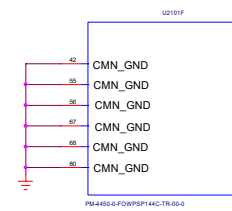
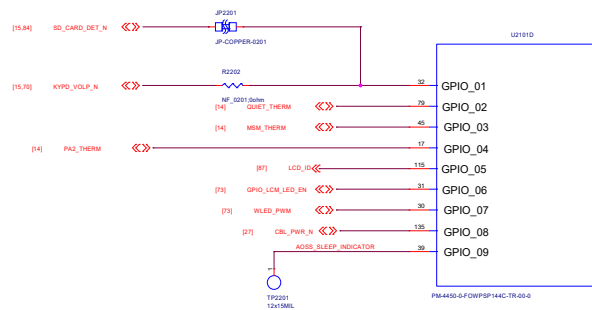
VDD_A_EBI
CH0 & CH1



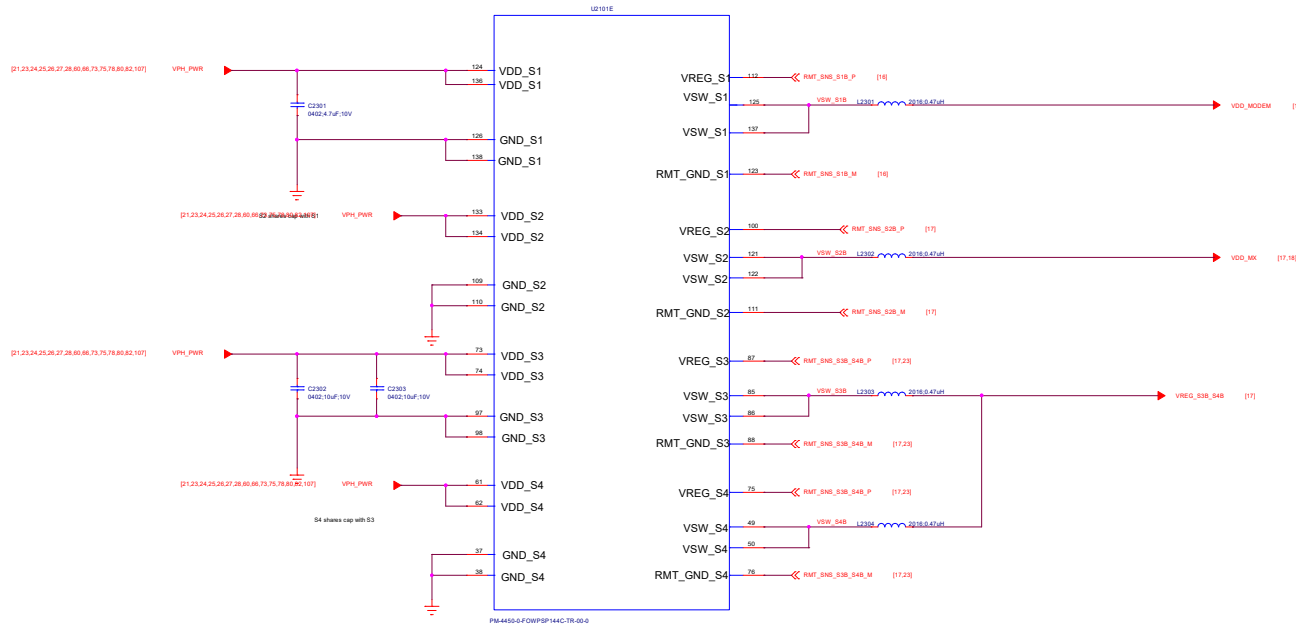




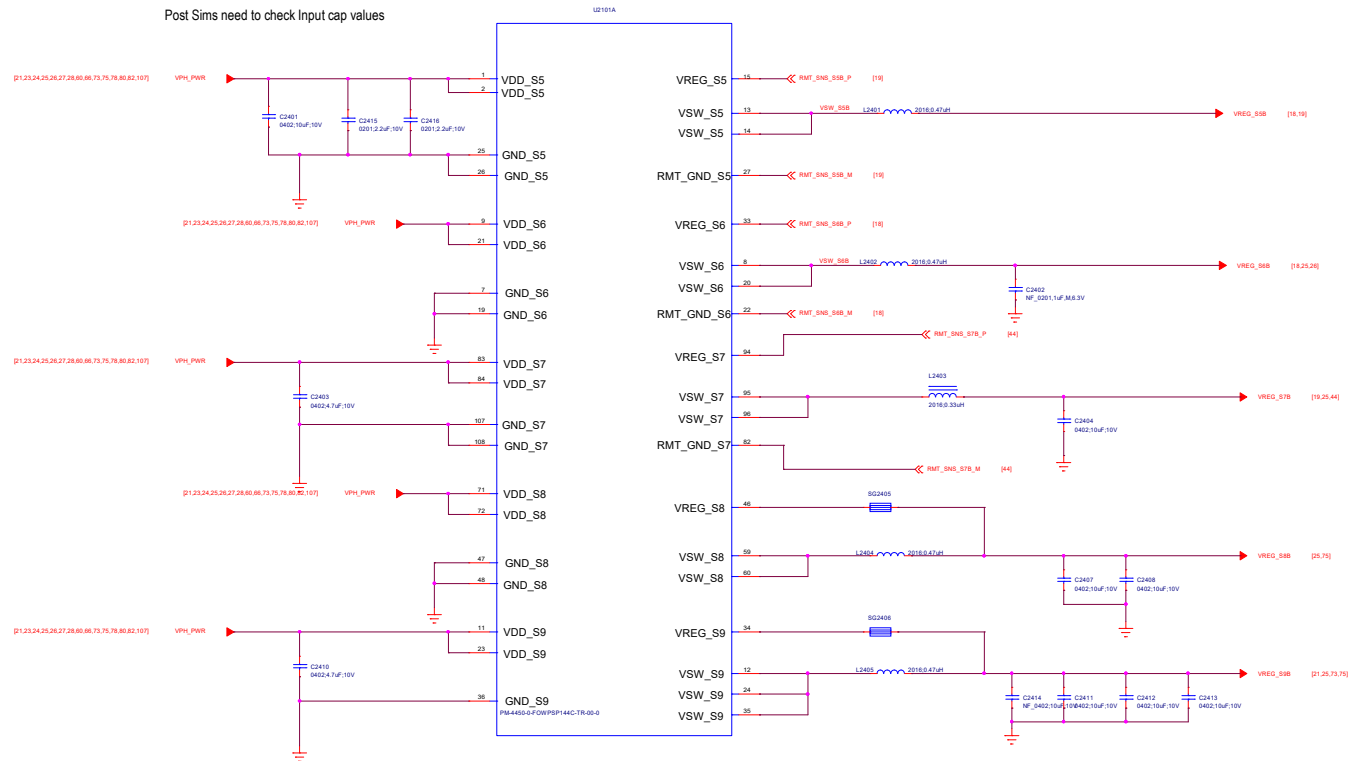
022_PM4450_GND_GPIO



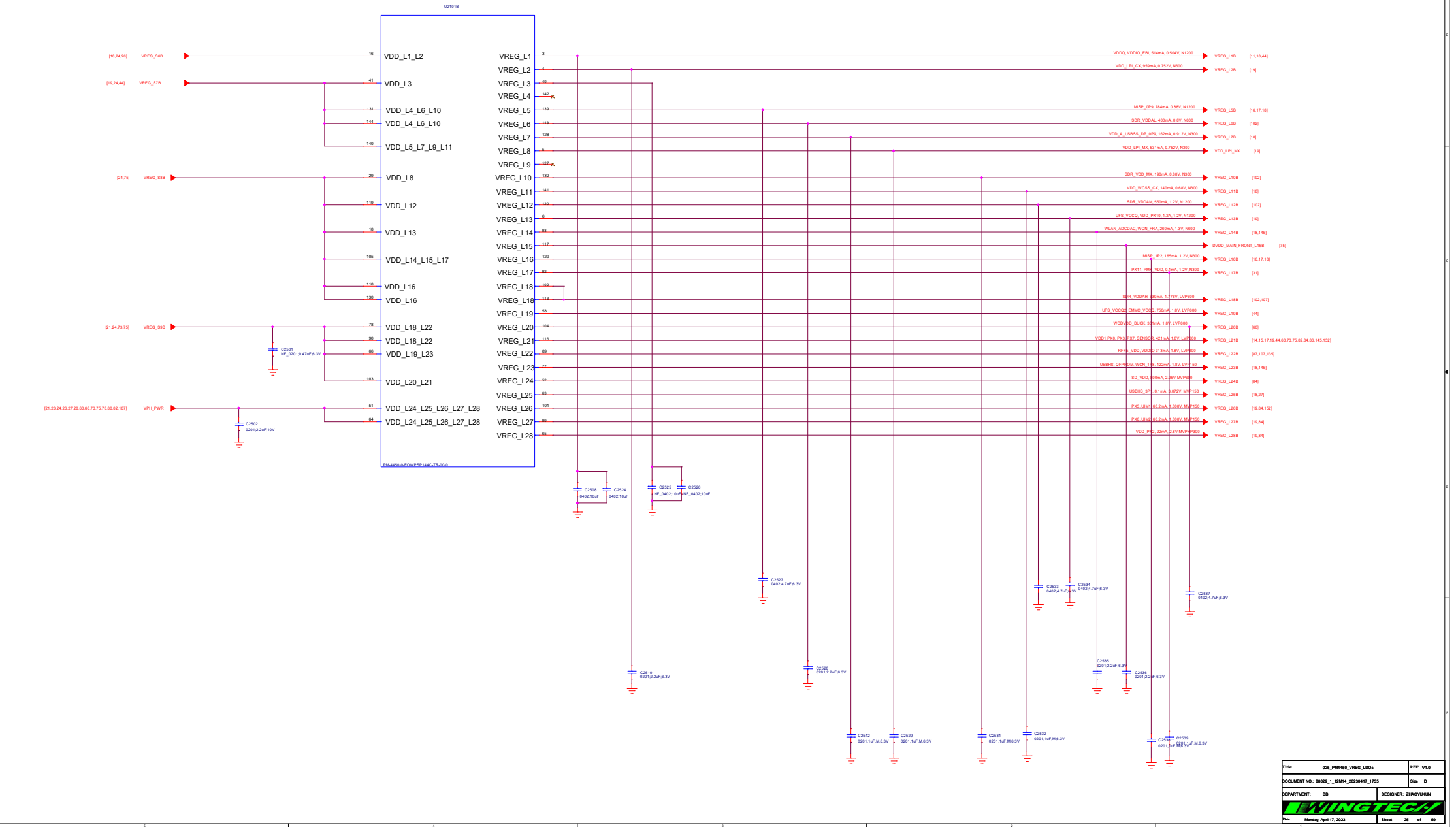
023_PM4450_VREG_S1_S4

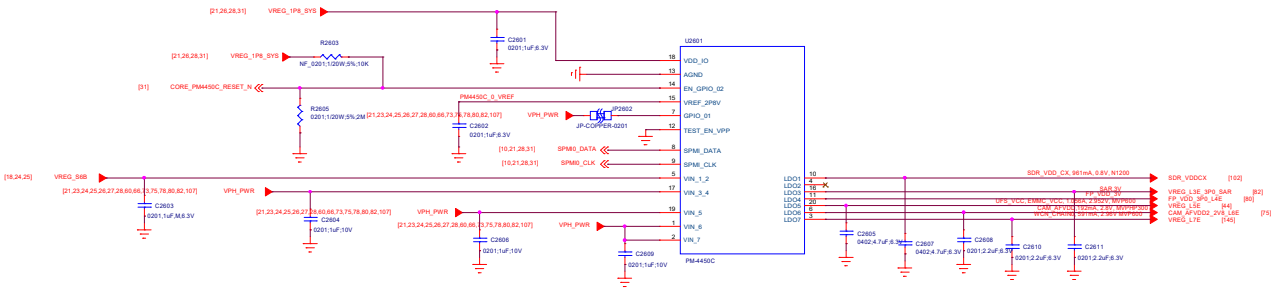


The Buck feedback lines must e shorted under the PMIC for multi-phase bucks.
The reference layout will not have this short under the PMIC; a generic breakout.

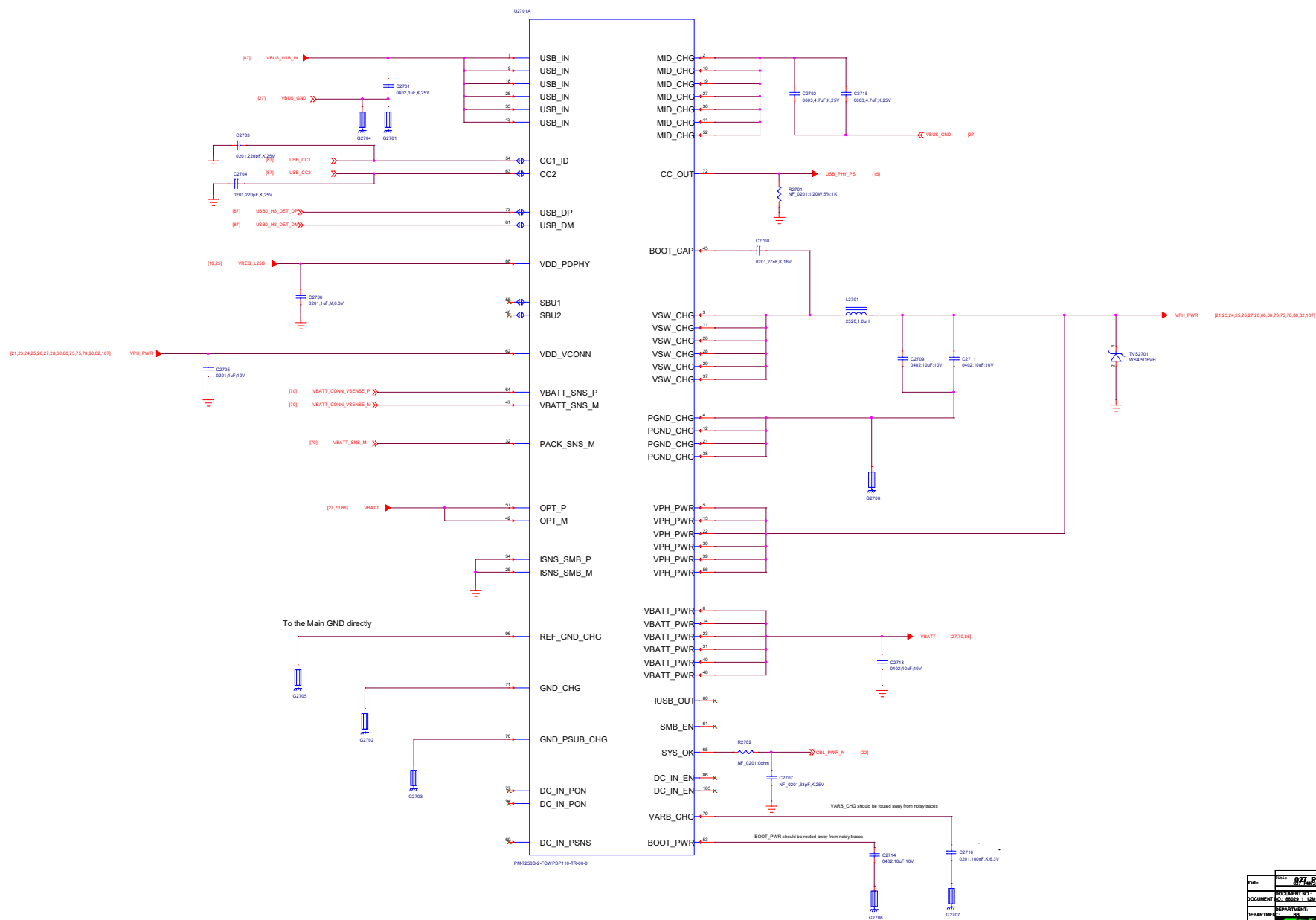


025_PM4450_VREG_LDOs

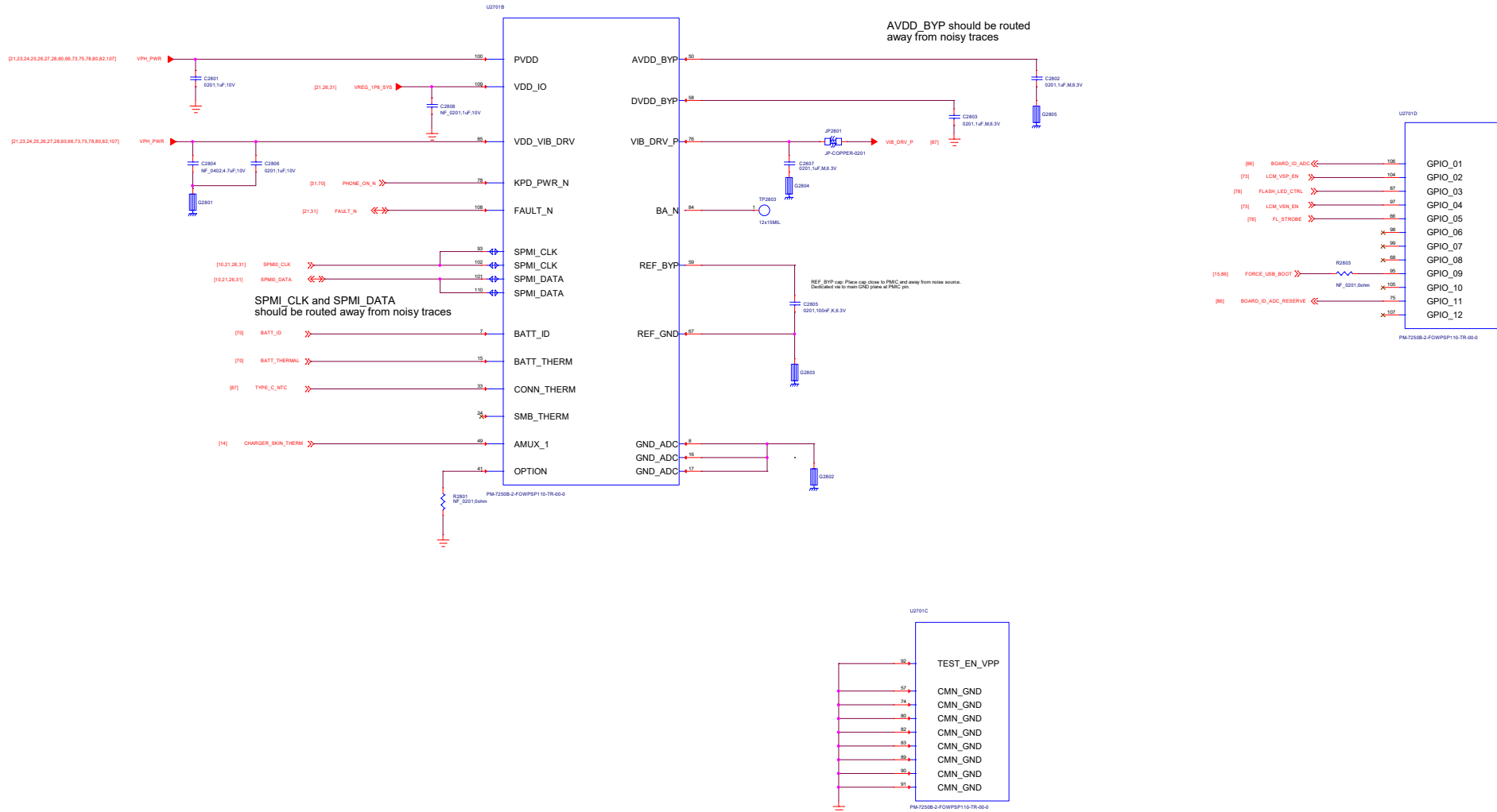




027_PMIC_PM7250B_TIOMAN: CHARGE

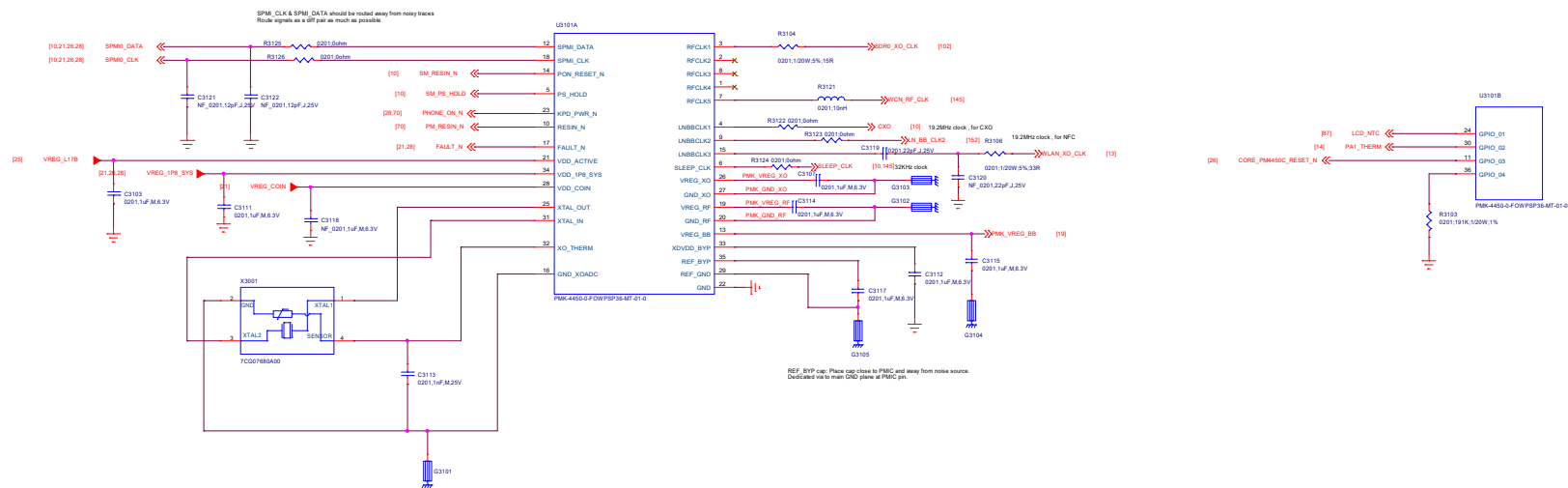


028_PMIC_PM7250B_TIOMAN: MISC/GPIO/GNDE



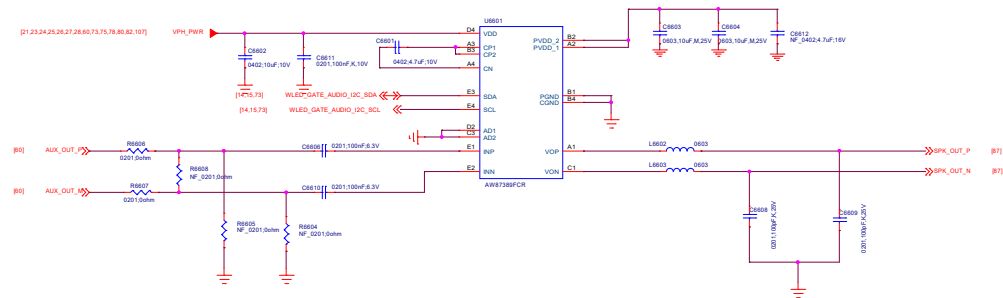
File	028_PMI7250B_TIOMAN: MISC/GPIO/GNDE	Rev	1.0
Document	028_PMI7250B_TIOMAN: MISC/GPIO/GNDE	Rev	1.0
Department	SS	Designed	SS
Check	SS	Checked	SS
Date	2024-04-17-2025	Sheet	40 of 50

031_PMK4450



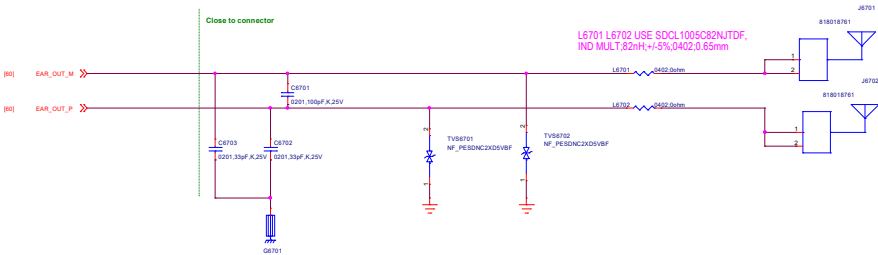
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DOCUMENT NO.: 68529_1_12M14_20230417_1755	Size D	
DEPARTMENT: BB	DESIGNER: ZHAOYUKUN	
		
Date: Monday, April 17, 2023	Sheet	31 of 50

066_SPK



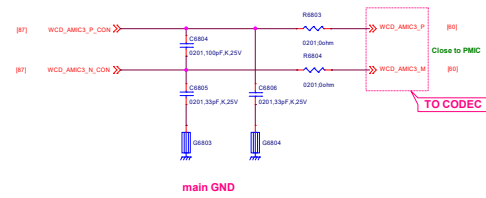
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DOCUMENT NO.:		88029_1_12M14_28230417_1755	Size D
DEPARTMENT:		88	DESIGNER: ZHAOYUKUN
			
Date:		Monday, April 17, 2025	Sheet 06 of 06

067_Receiver

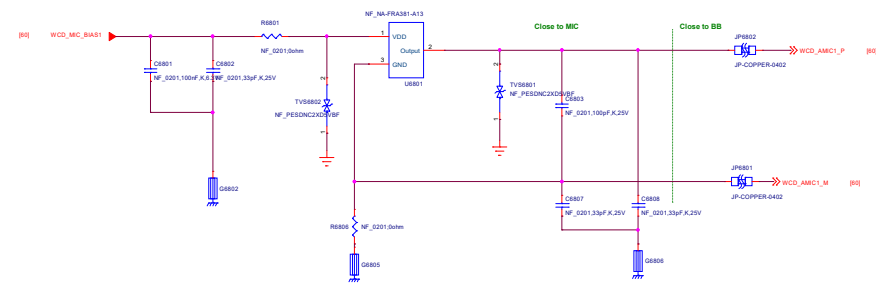


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DOCUMENT NO.: 86029_1_13M14_20230417_1755		Size: D
DEPARTMENT: BB		DESIGNER: ZHAOYUKUN
		
Date: Monday, April 17, 2023		Sheet: 67 of 58

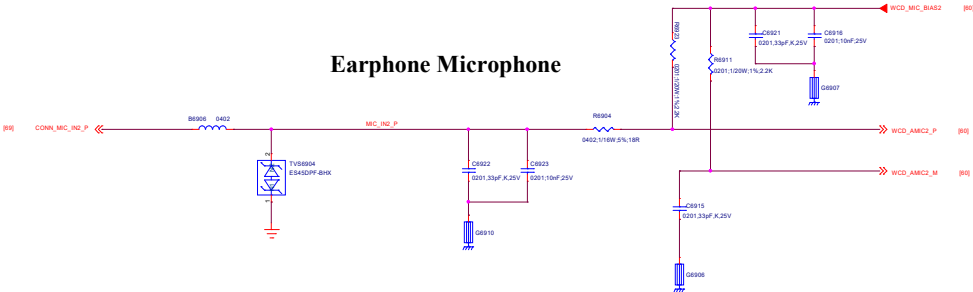
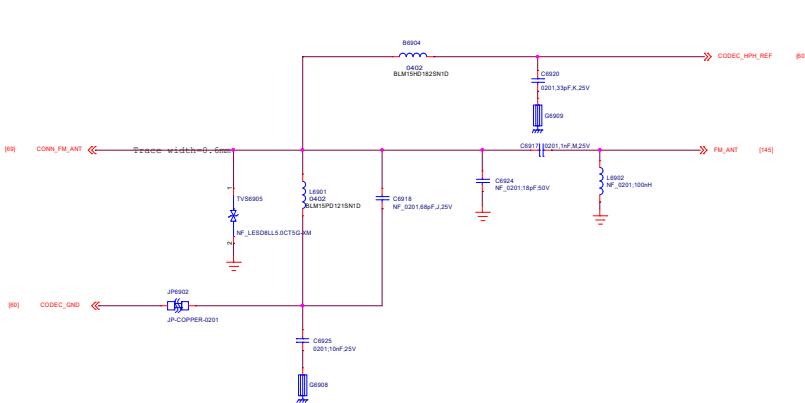
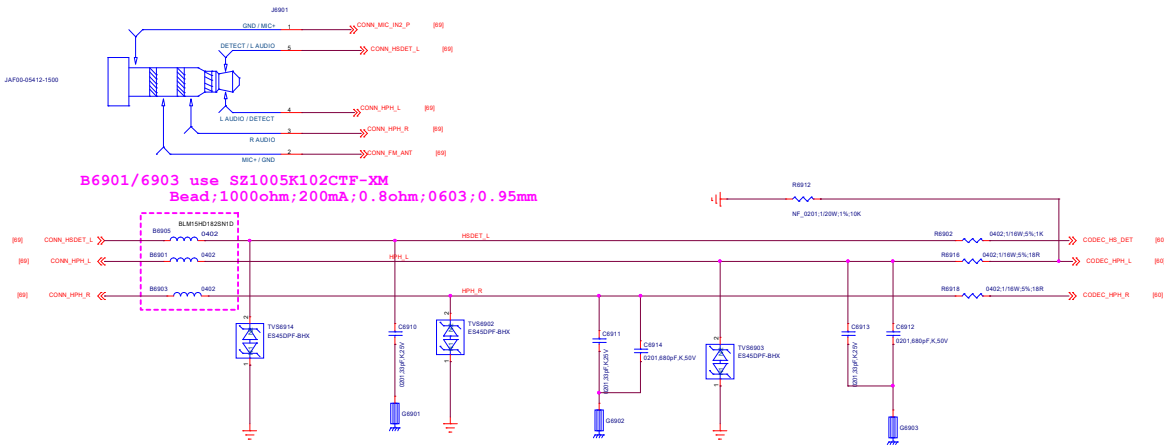
MAIN MIC



RFI兼容设计 Second MIC



069_EARPHONE

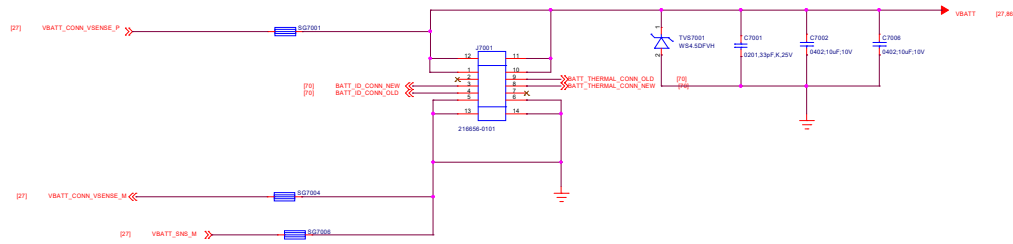


Earphone Microphone

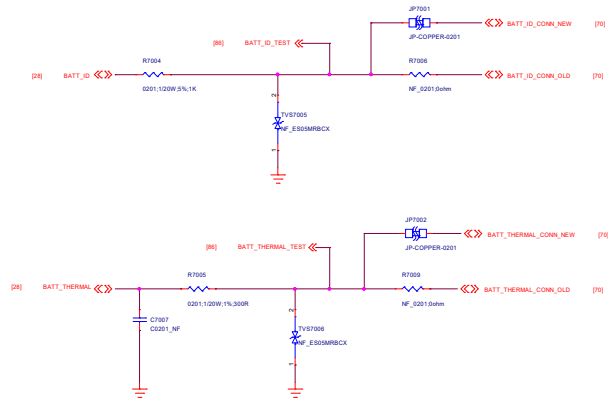
070_Battery/Sidekey

BATTERY CONNECTOR

BATTERY MASTER CONNECTOR



BATTERY ID&THERMAL

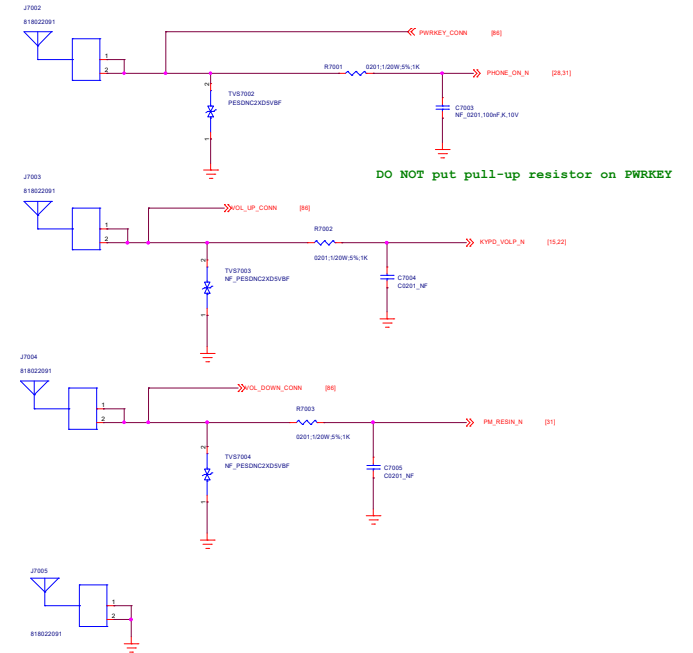


Power Key / Key Pad

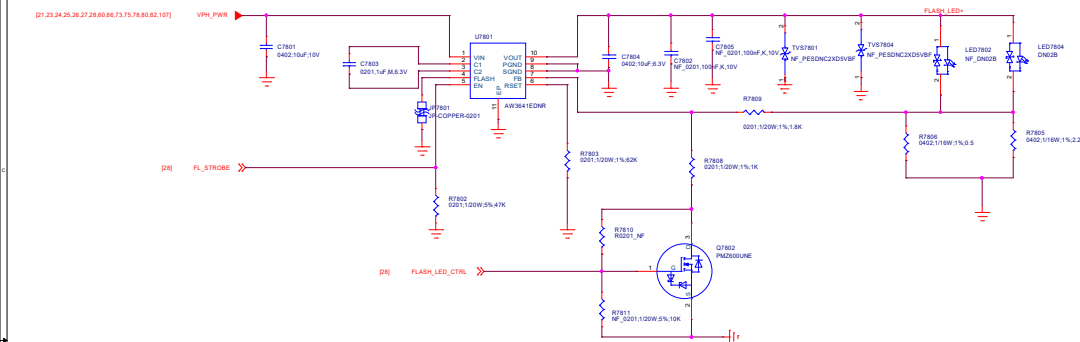
Power Key

Power Up

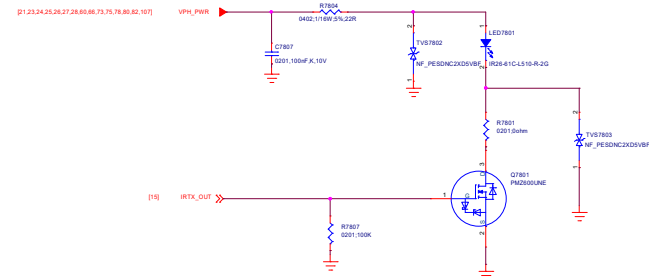
Power Down

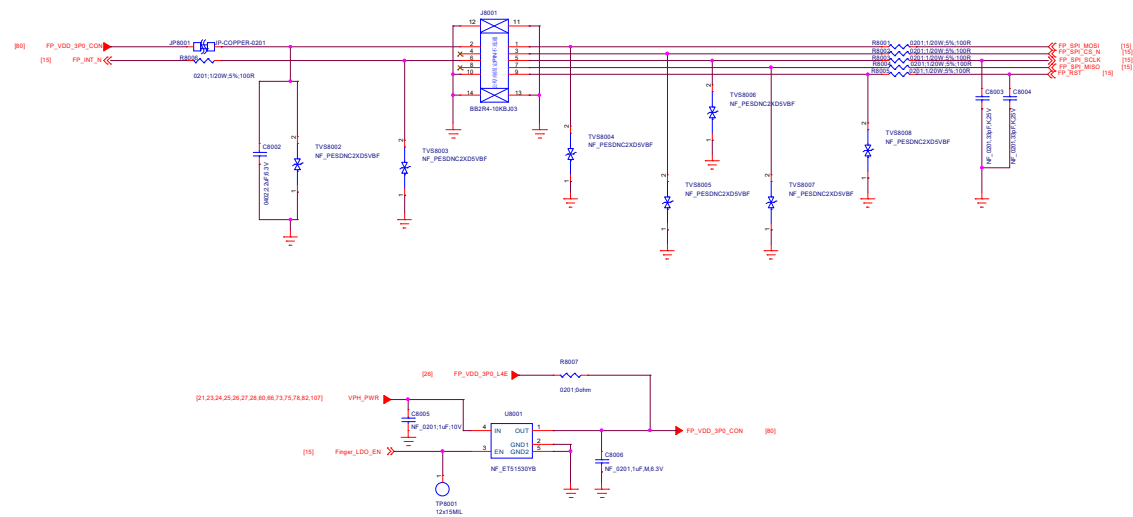


Flash LED



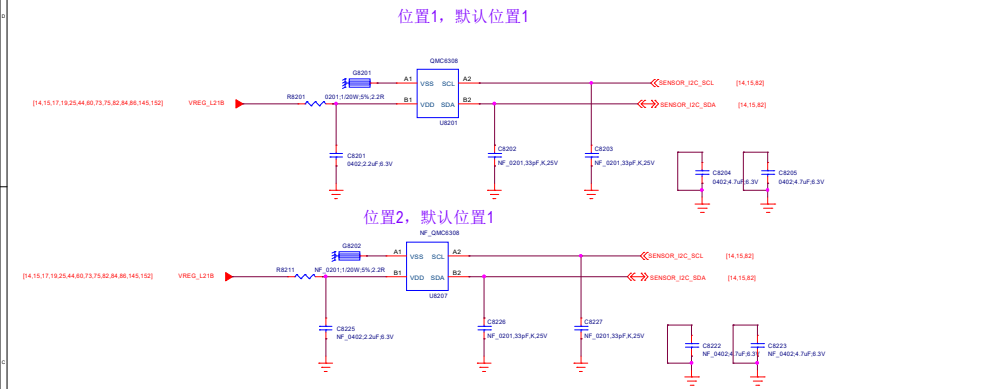
IR LED



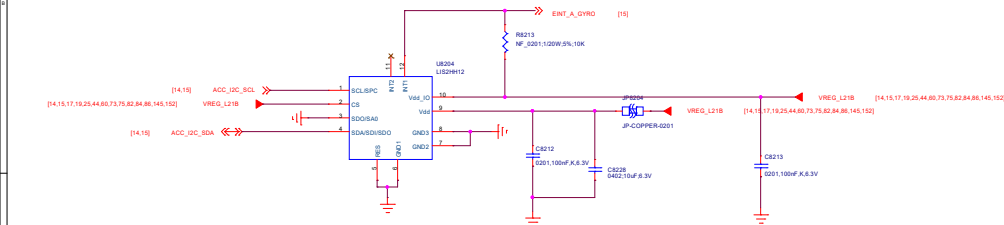


082_SENSORS

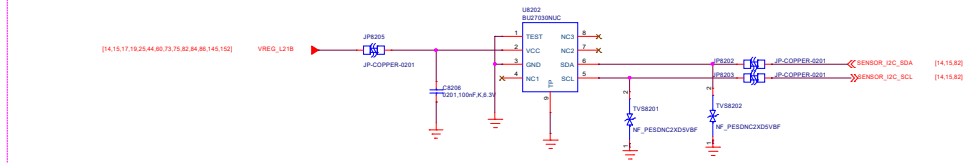
M-SENSOR



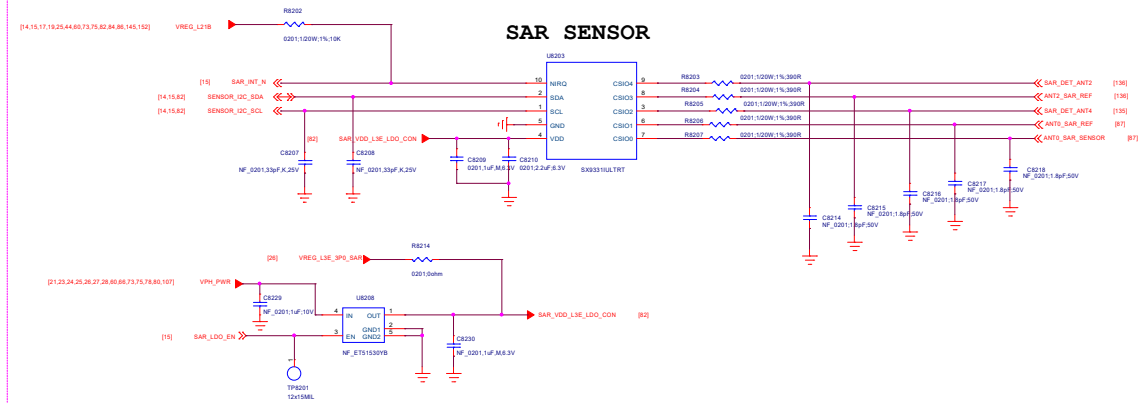
Accerometer



FRONT ALS SENSOR

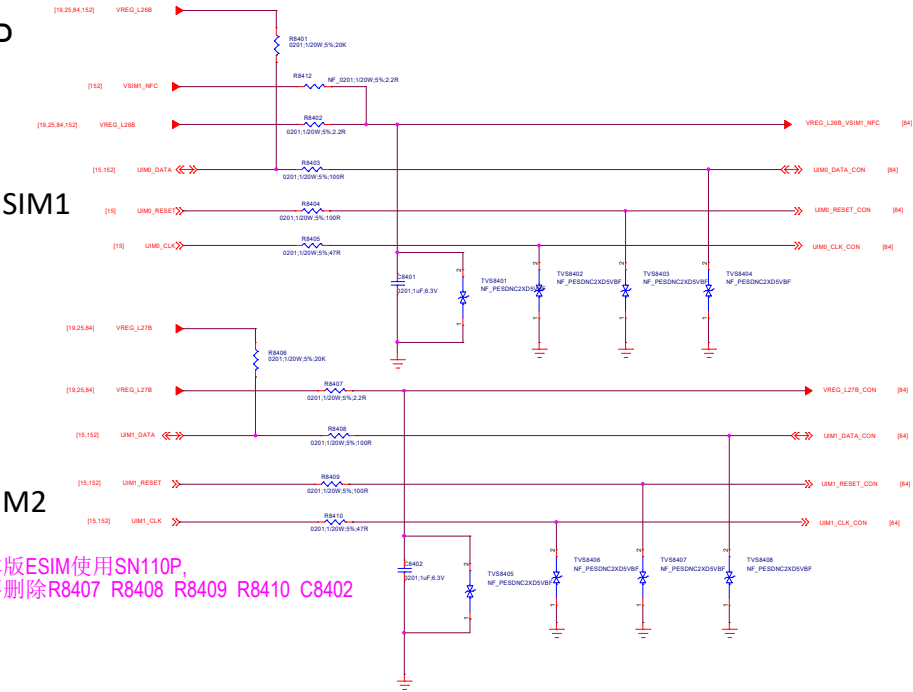


SAR SENSOR



084_SIM/TF IF

SIM CARD

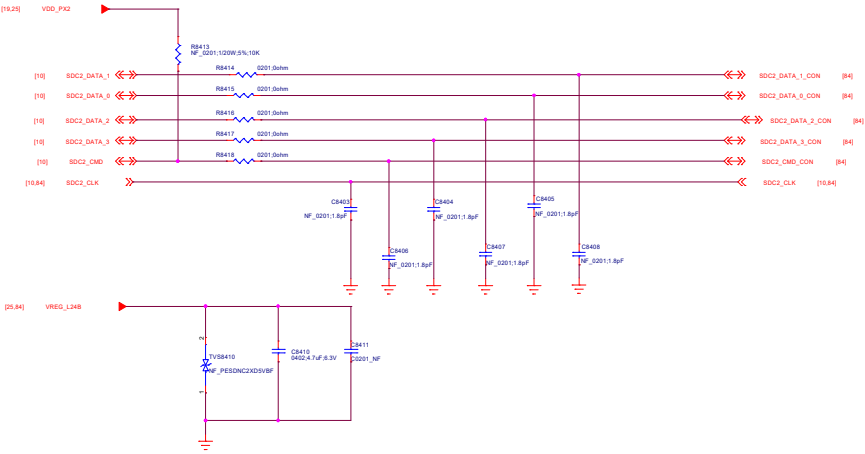


SIM1

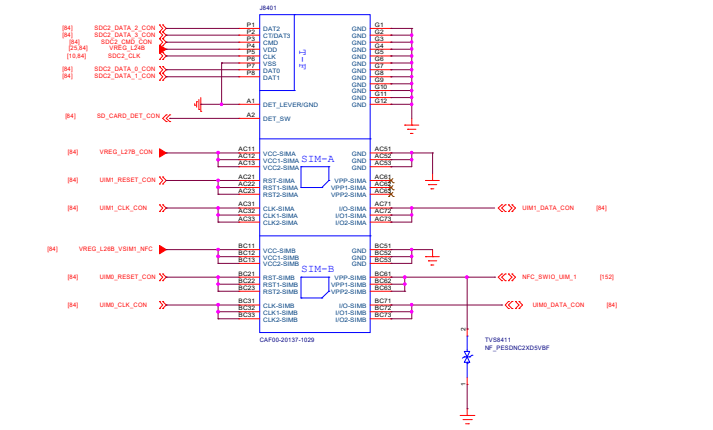
SIM2

日本版ESIM使用SN110P,
需要删除R8407 R8408 R8409 R8410 C8402

T_Card

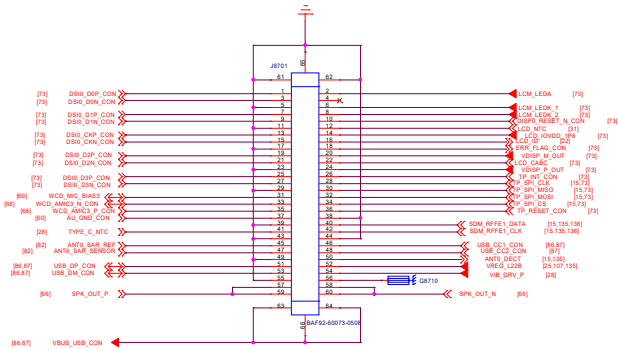


DETECT

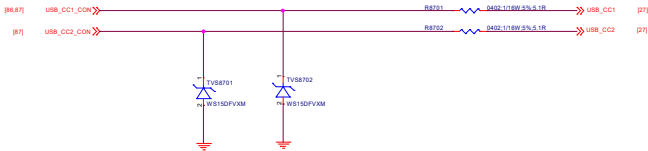


087_Sub PCB IF

CONNECTOR

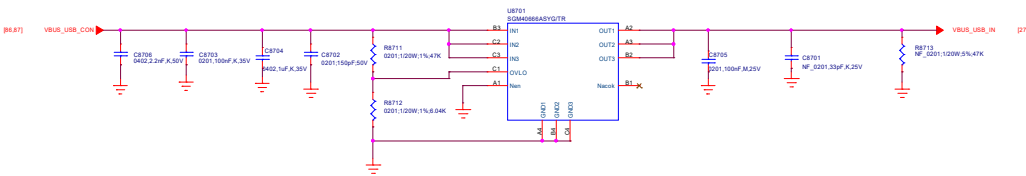


CC

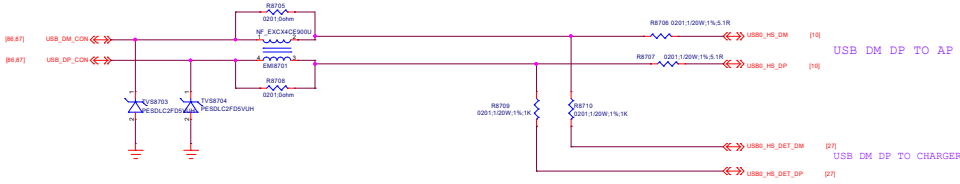


OVP

OVP_VBUS



USB IF



102.RF_Transceiver_SDR435 WMSS

SDR_VDDCX Max current (mA)
VDD_CX1 320 mA
VDD_CX2 320 mA
VDD_CX3 320 mA

VREG_L10B Max current (mA)
VDD_MX0 12 mA
VDD_MX1 9 mA
VDD_MX2 87 mA
VDD_PHY 83 mA

VIO18_RFFE Max current (mA)
VDD_PX 10 mA

VREG_L6B Max current (mA)
VDDAL_DIG_1 19.75 mA
VDDAL_DIG_2 9.25 mA
VDDAL_DIG_3 12.5 mA
VDDAL_DIG_4 12.5 mA
VDDAL_GNSS 7.5 mA
VDDAL_RX_A 75.6 mA
VDDAL_RX_B 57.9 mA
VDDAL_RX_0 50 mA
VDDAL_RX_1 50 mA
VDDAL_RX_2 50 mA
VDDAL_TX0_1 52.25 mA
VDDAL_TX1_1 52.25 mA
VDDAL_XO 12 mA

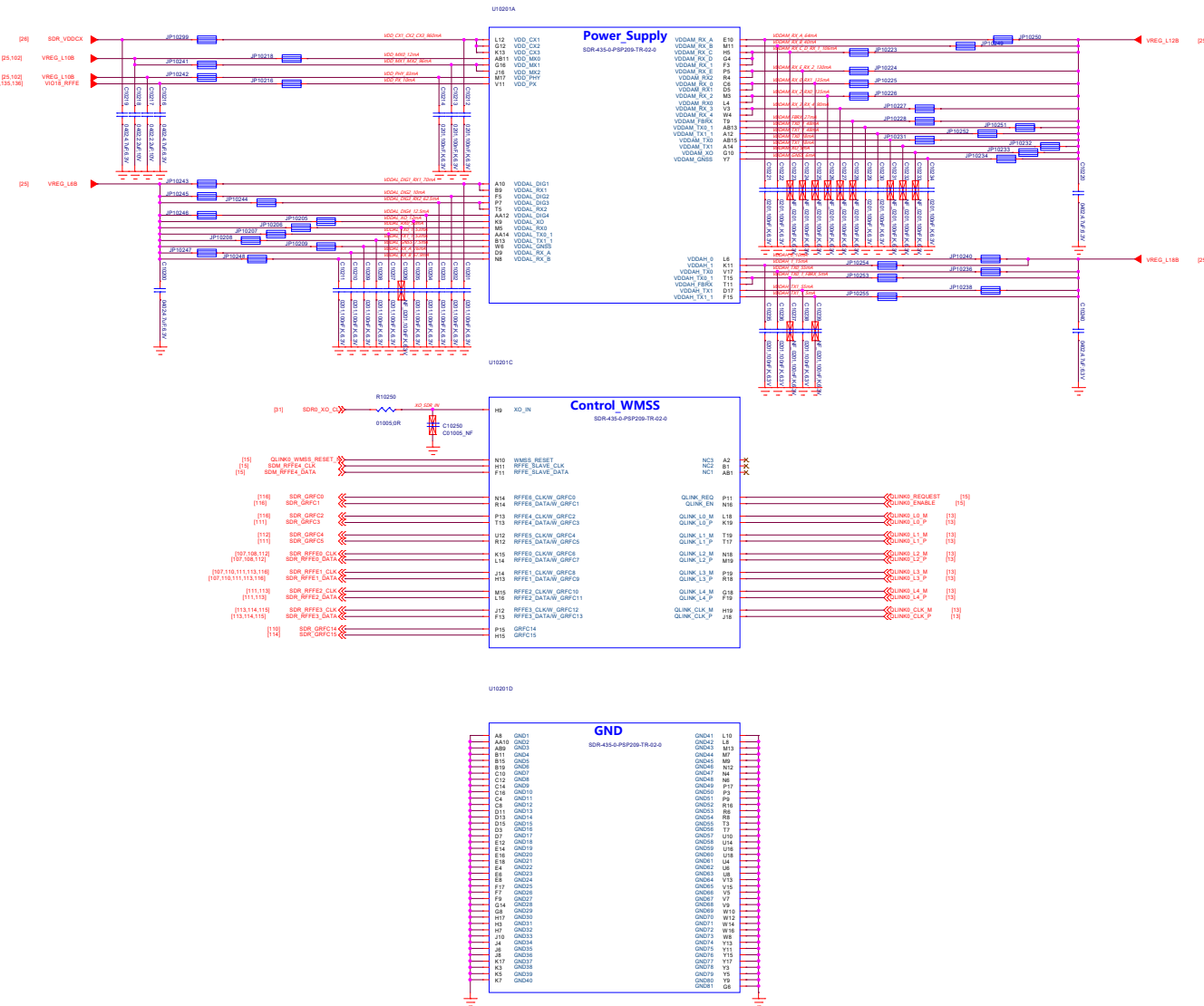
REVISIONS		
LTR	DESCRIPTION	DATE

VREG_L12B Max current (mA)

VDDAM_FBRX 27 mA
VDDAM_GNSS 6 mA
VDDAM_RX_0 30 mA
VDDAM_RX_1 60 mA
VDDAM_RX_2 30 mA
VDDAM_RX_3 60 mA
VDDAM_RX_4 30 mA
VDDAM_RX_A 64 mA
VDDAM_RX_B 40 mA
VDDAM_RX_C 23 mA
VDDAM_RX_D 23 mA
VDDAM_RX_E 23 mA
VDDAM_RX0 105 mA
VDDAM_RX1 105 mA
VDDAM_RX2 107 mA

VREG_L18B Max current (mA)

VDDAH_0 10 mA
VDDAH_FBRX 15 mA
VDDAH_1 15 mA
VDDAH_TX0 55 mA
VDDAH_TX0_1 5 mA
VDDAH_TX1 55 mA
VDDAH_TX1_1 5 mA



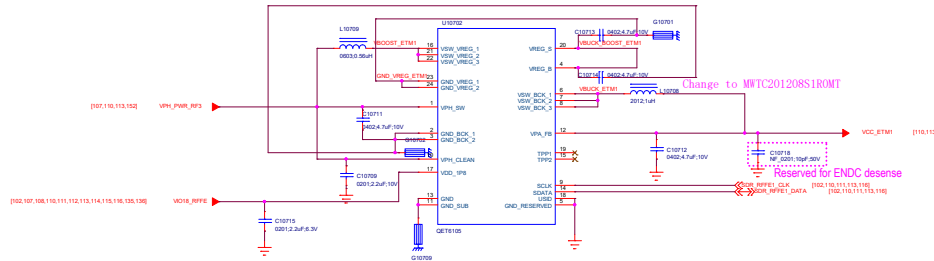
Title: 102.RF_Transceiver_SDR435 WMSS v1.0	
Document No: 0020_1_1004_0030417_1706	Rev: 0
Department: WINGTECH SA	Designer: Xun Wu
Date: Monday, April 17, 2023	
Sheet 102 of 158	

107.APT/ET_QET6105

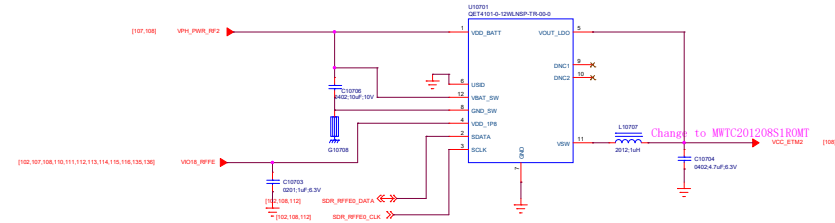
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LTR	DESCRIPTION	DATE	APPROVED
A	INITIAL RELEASE		



QET6105_CH1



QET4101_CH0

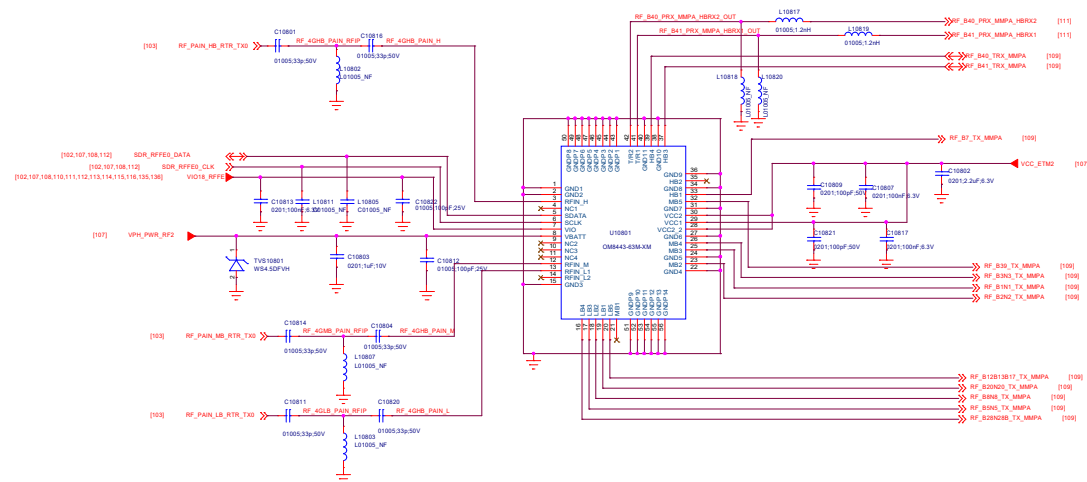


Title: 107.APT_Module		REV: V02
DOCUMENT NO.: 00023_1_130614_0020047_1700		Rev: Custom
DEPARTMENT: VINGTECH-NA		DESIGNER: Sun Vial
Date: Monday, April 17, 2005		Sheet: 107 of 108

VERSION	1	EDITED BY	LAST_EDITOR	LAST EDIT DATE	2-23-2007_10:06
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108MMPA_Phase5N#0

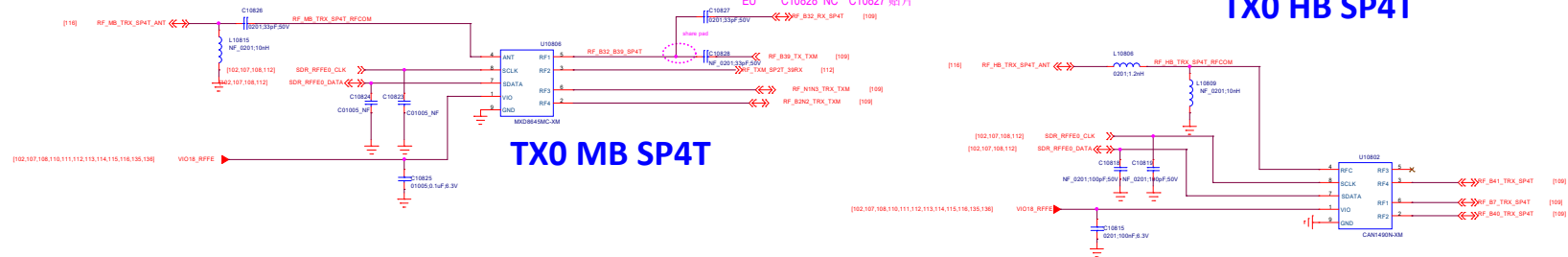
MMPA for NR FDD



Notice: key component U10802 and U10806 should be different due to the same MIP1

JP/Asia	C10828 贴片	C10827 NC
EU	C10828 NC	C10827 贴片

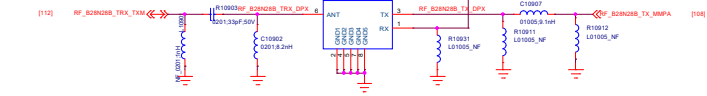
TX0 HB SP4T



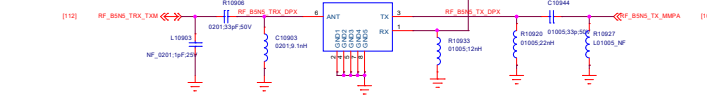
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DOCUMENT NO.: Design Name = 88829_8_12M14_2023041		JAMES D
DEPARTMENT: DEPARTMENT - RF_Notes	DESIGNER: DESIGNER - XA	
		
Date: Page Modify Date = Monday, April 17, 2023	Sheet 108 of 156	

109.Phase5N#0_DUP

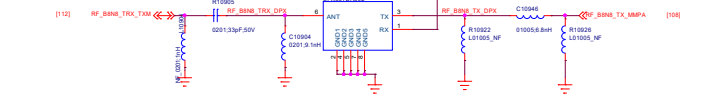
N28B28 TRX



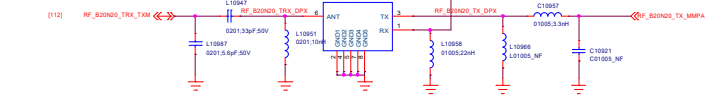
B26N5 TRX



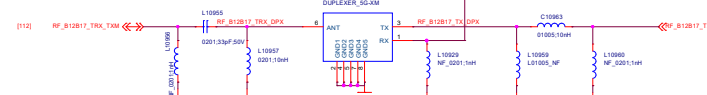
B8N8 TRX



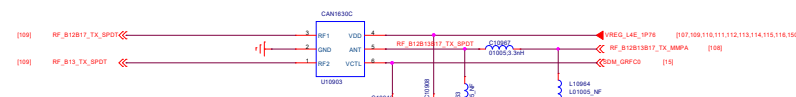
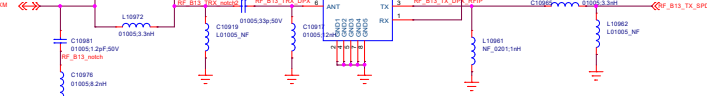
B20 N20 TRX



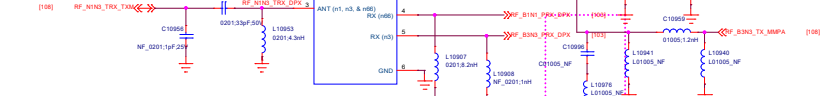
B12/17 TRX



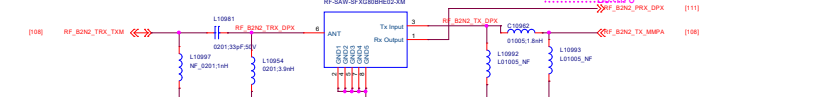
B13 TRX



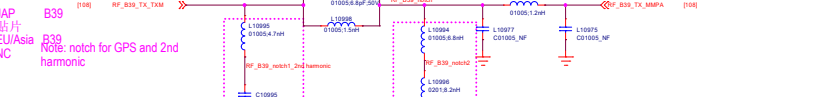
N1+N3 TRX



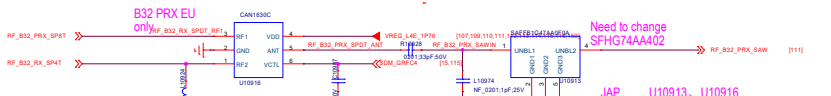
B2 TRX



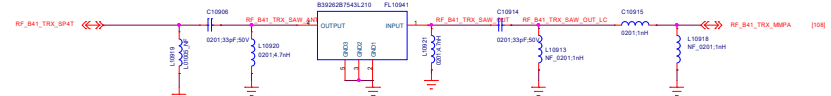
B39 TX



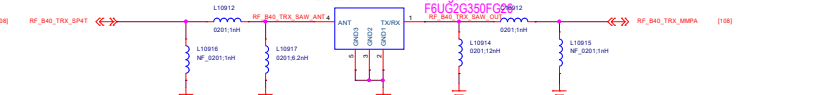
B32 RX



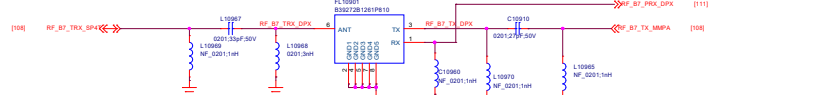
B41 TRX



B40 TRX



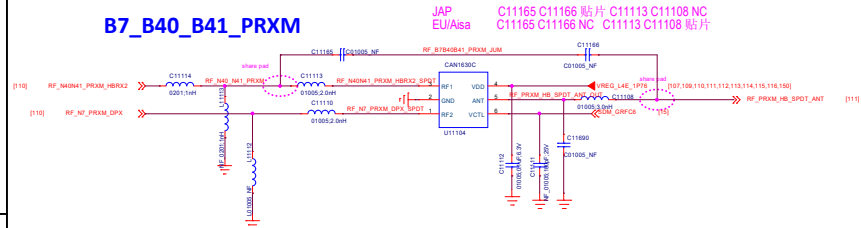
B7 TRX



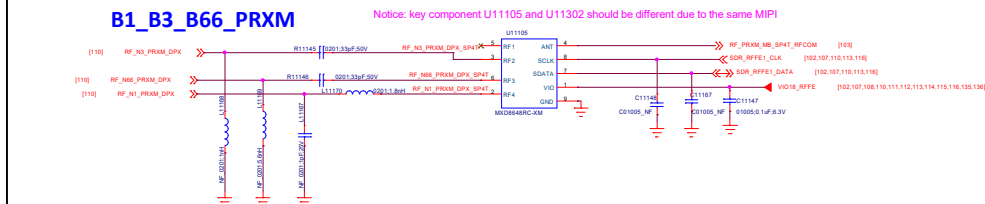
Title:	Rev: V00
DOCUMENT NO.: Design Name - 00009_1_13ME4_2023041	0000 D
DEPARTMENT: DEPARTMENT - RF_2nd	DESIGNER: XIAN
DATE: 2023.04.11	Sheet 001 of 010

111.PRX PRXM

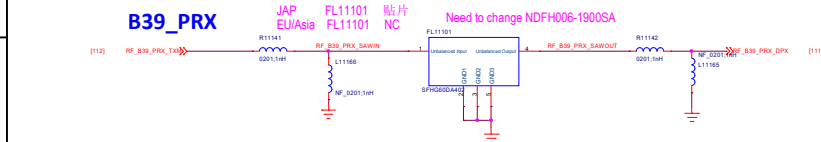
B7_B40_B41_PRXM



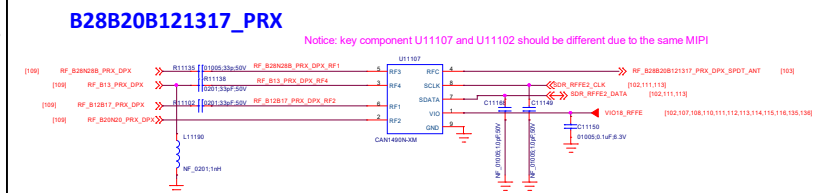
B1_B3_B66_PRXM



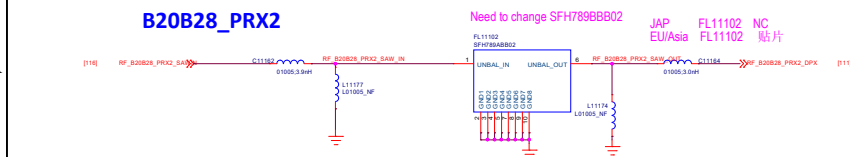
B39 PRX



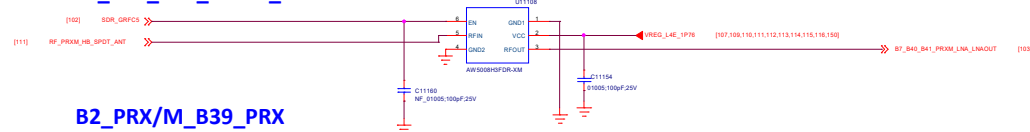
B28B20B121317_PRX



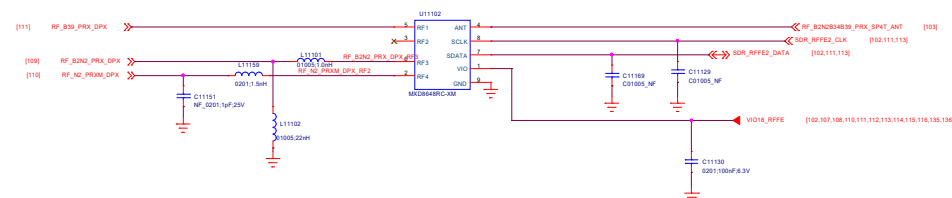
B20B28 PRX2



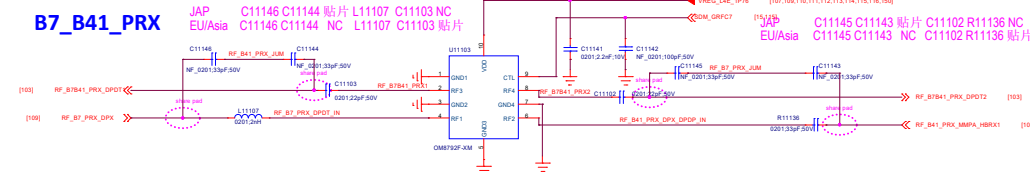
B7_B40_B41_PRXM_LNA



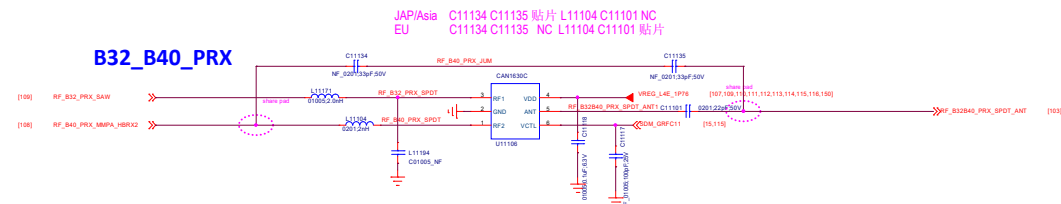
B2 PRX/M B39 PRX



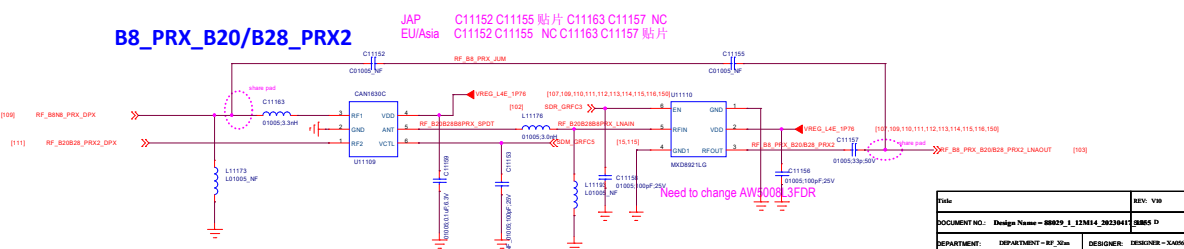
B7 B41 PRX



B32_B40_PRX

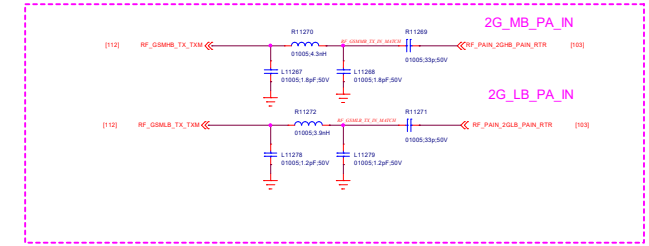
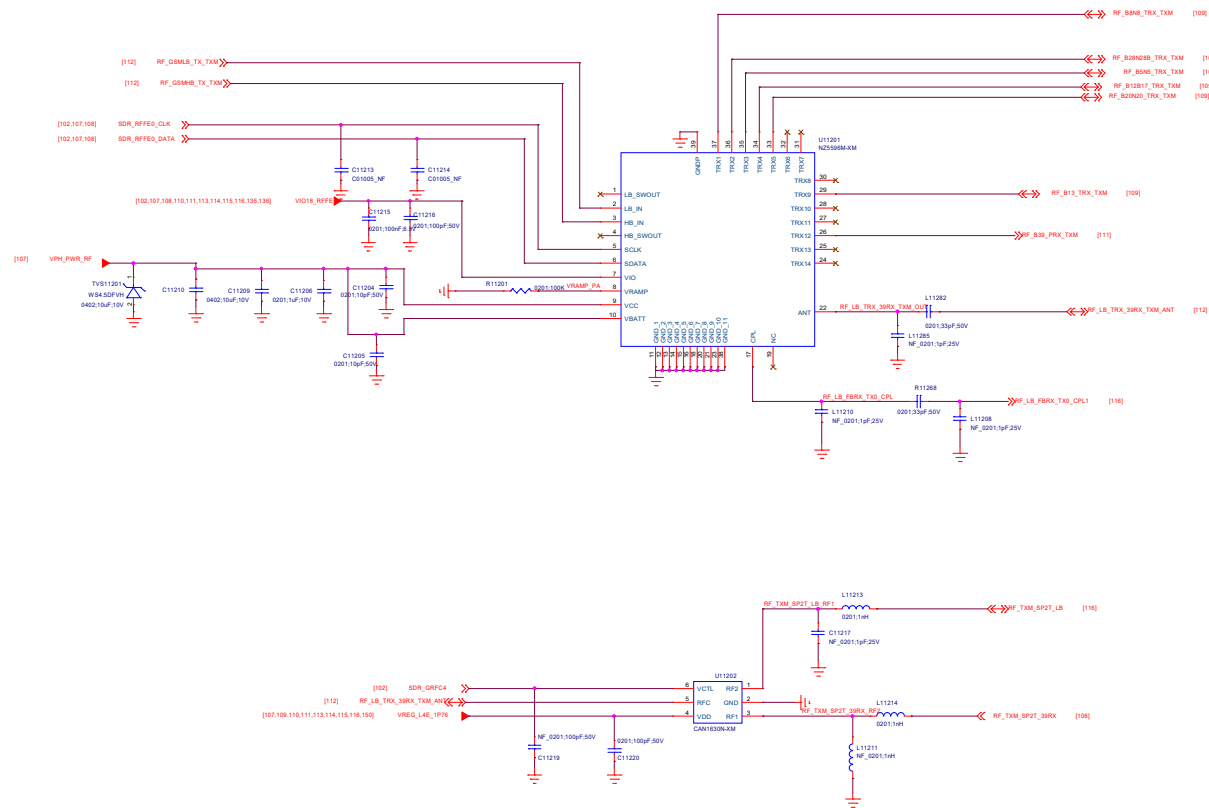


B8 PRX B20/B28 PRX2



Title		REV: V10
DOCUMENT NO.: Design Name = 80829_1_12M14_20230411		80865 D
DEPARTMENT: DEPARTMENT - RF_20m	DESIGNER: DESIGNER - XA050	
		
Date: Page	Modify Date: Monday, April 17, 2023	Sheet 111 of 156

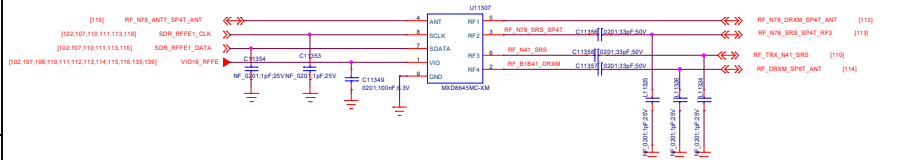
112.TXM_VC7916-36M



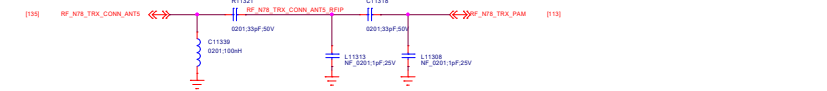
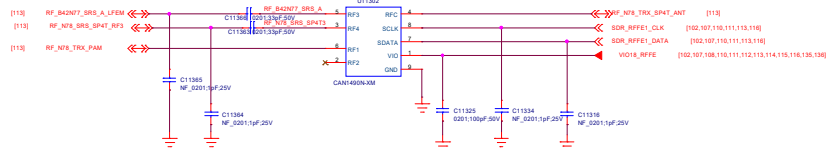
Title		REV: V10
DOCUMENT NO.: Design Name = 88029_1_13M14_2023041		JB05 D
DEPARTMENT: DEPARTMENT - RF_20m	DESIGNER: JB0506	
		
Date:	Page Modify Date = Monday, April 17, 2023	Sheet 112 of 156

113.N77_N78_PAM

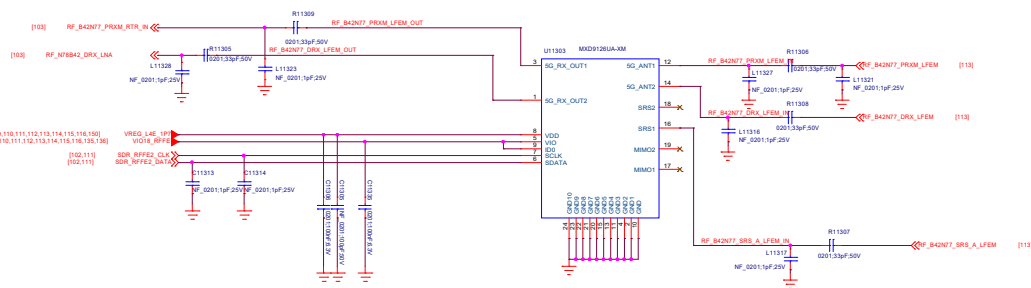
ANT7 SP4T



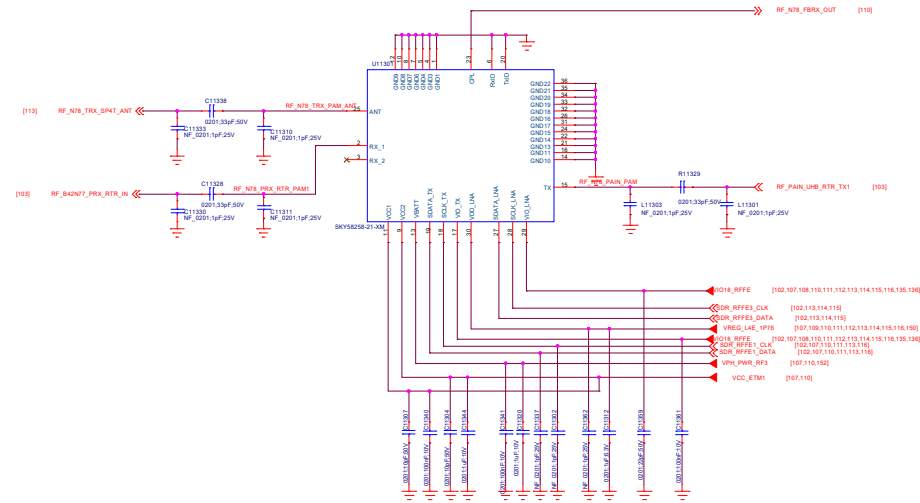
N78 SRS SP4T



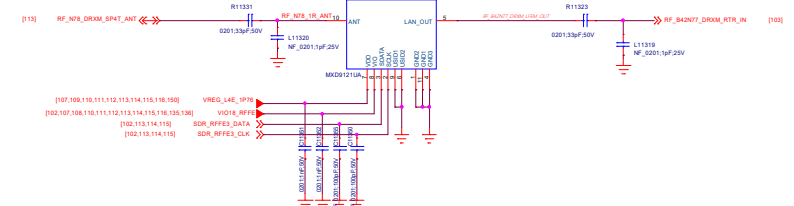
N78 LFEM PRXM/DRX



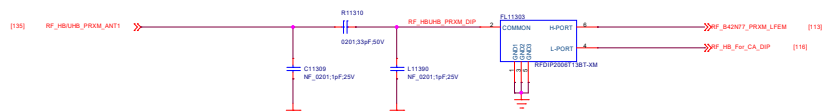
N78 PA



N78 LFEM DRXM

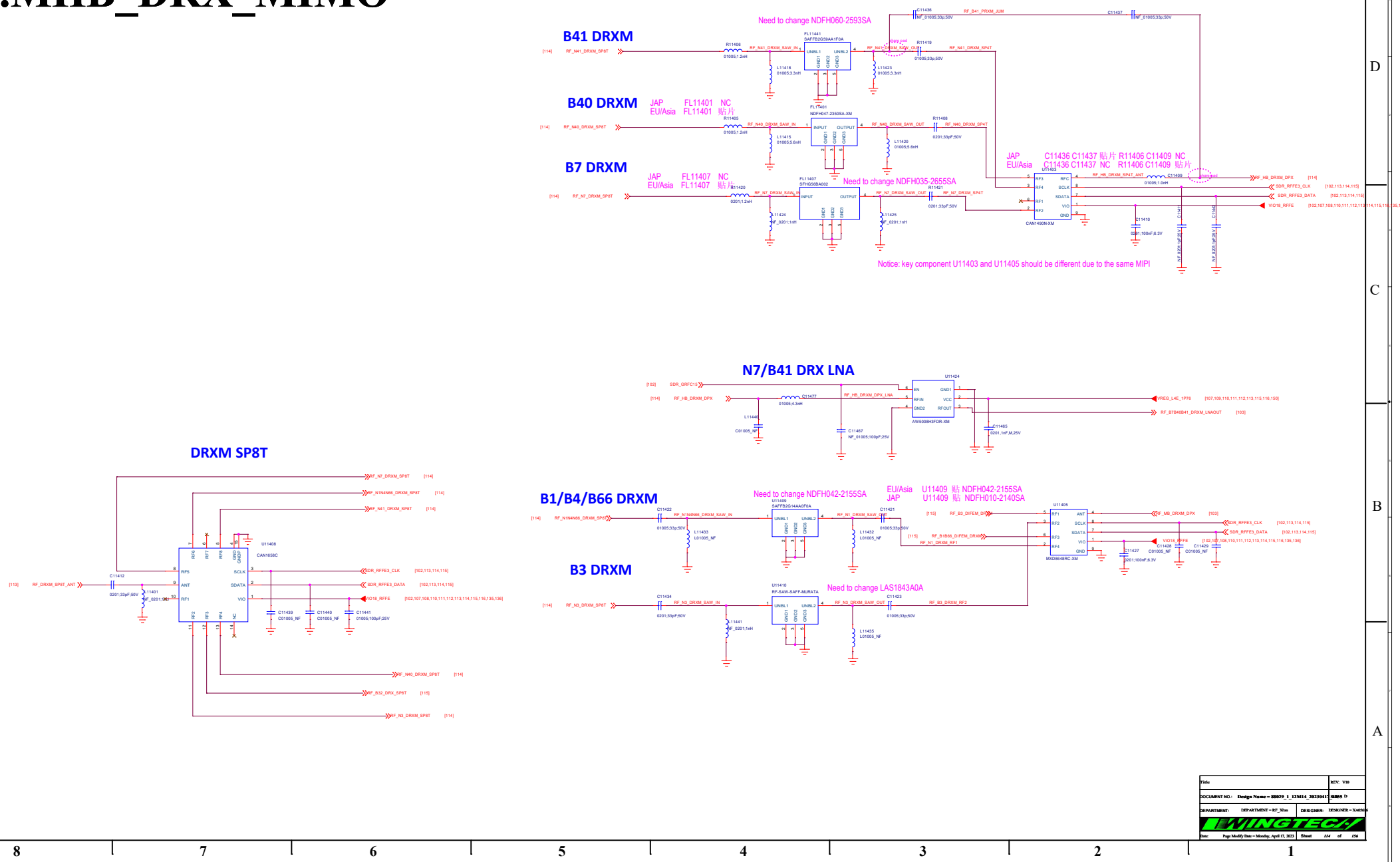


ANT1 N78+B41 DIP



Title:		REV: V08
DOCUMENT NO.: Design Name - 113N77_113N78_113N79		DESIGNER: XIAOYU
DEPARTMENT: DEPARTMENT - RF_3nm		DESIGNER: XIAOYU
Date: 2023-04-11		Sheet: 223 of 228

114.MHB_DRX_MIMO

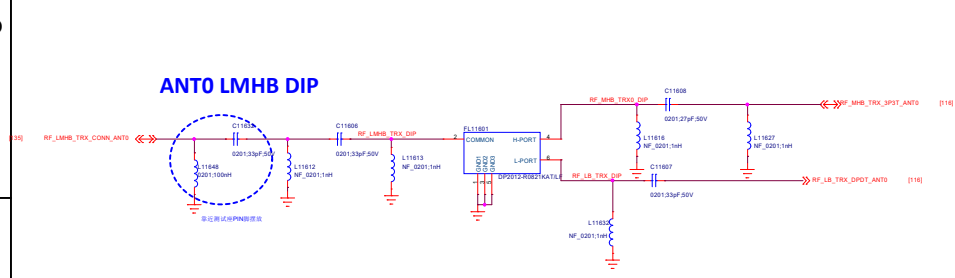


8	7	6	5	4	3	2	1
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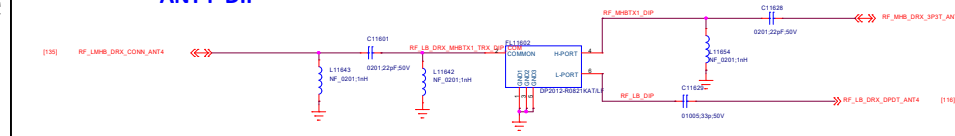


116.ASDIV_FBRX

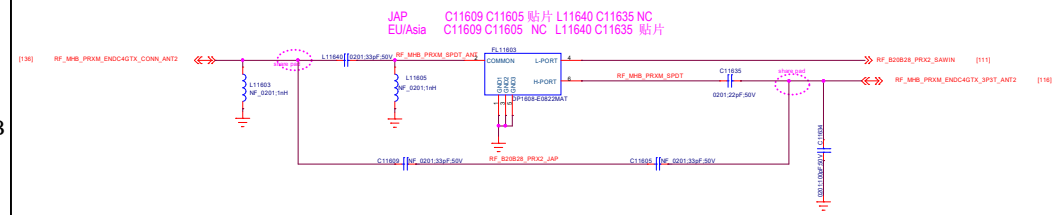
ANT0 LMHB DIP



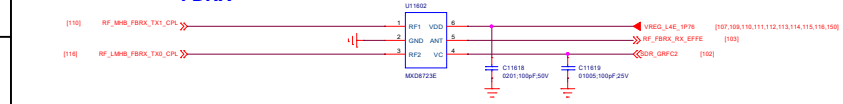
ANT4 DIP



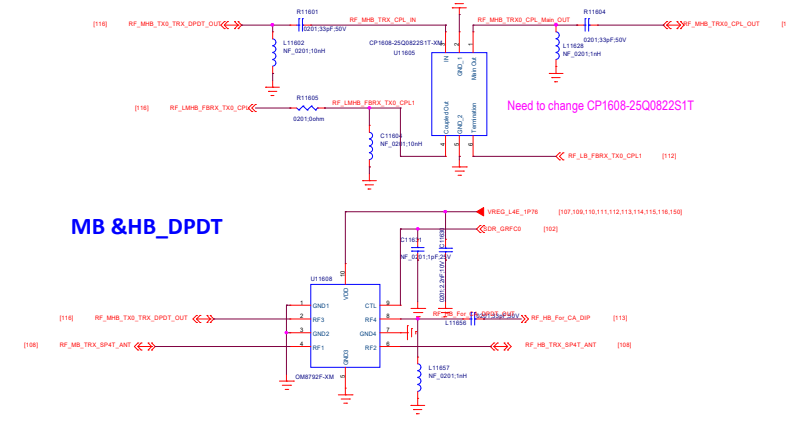
ANT2 MHB PRXM



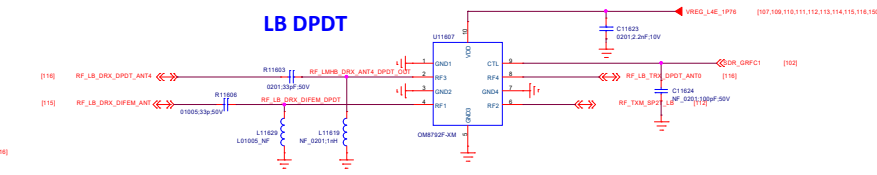
FBRX



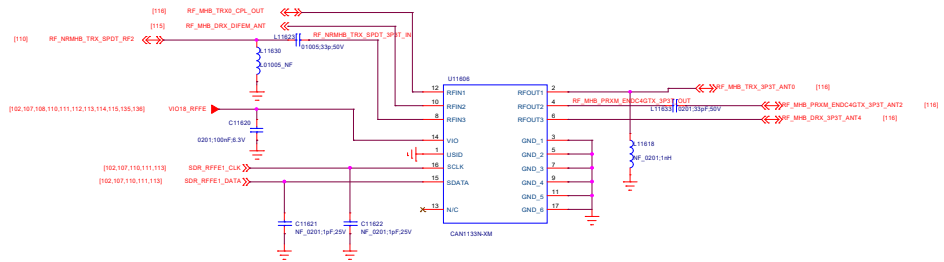
MB & HB_DPDT



LB DPDT



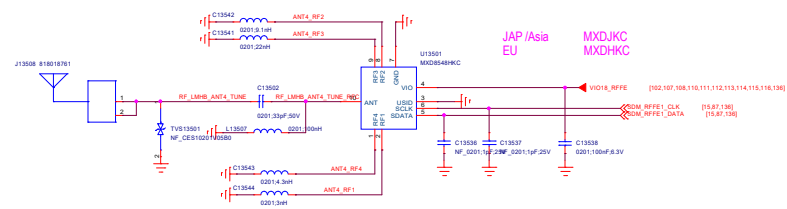
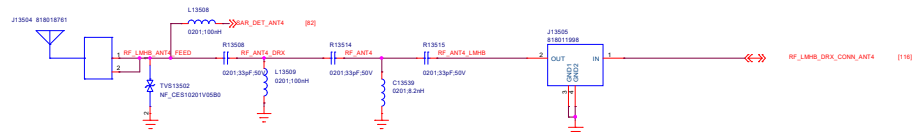
ANT0&ANT4_3P3T



135.WWAN_ANT_PART1

ANT4

DRX:4G/5G LB
PRXM:4G MHB
TRX:ENDC MHB

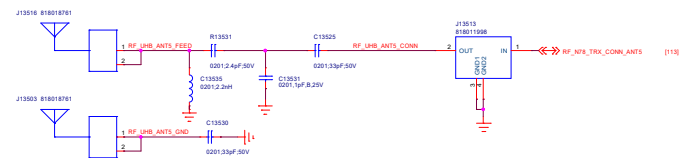


主板接地弹片



ANT5

N77/N78 PRI ANT



ANT0

N77/N78 PRI ANT

[15.87,107] VREQ_L32B R13532 [15.87] ANTS_DECT

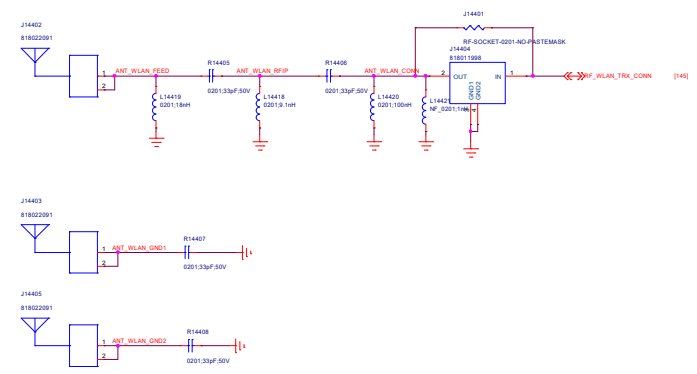


ANT1

N77/N78 PRI ANT



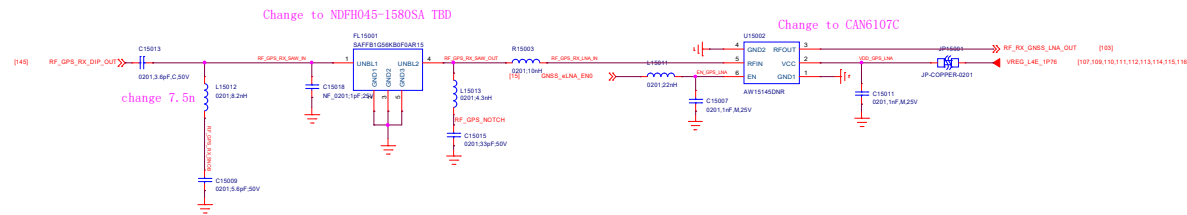
144.Connectivity_ANT



Title:		REV: V00
DOCUMENT NO.: Design Name - 88829_1_13ME4_2823841		8885 D
DEPARTMENT:	DEPARTMENT - RF_3m	DESIGNER: DESIGNER - XANXIN
Date: Page Modify Date - Monday, April 11, 2022		Sheet 244 of 258



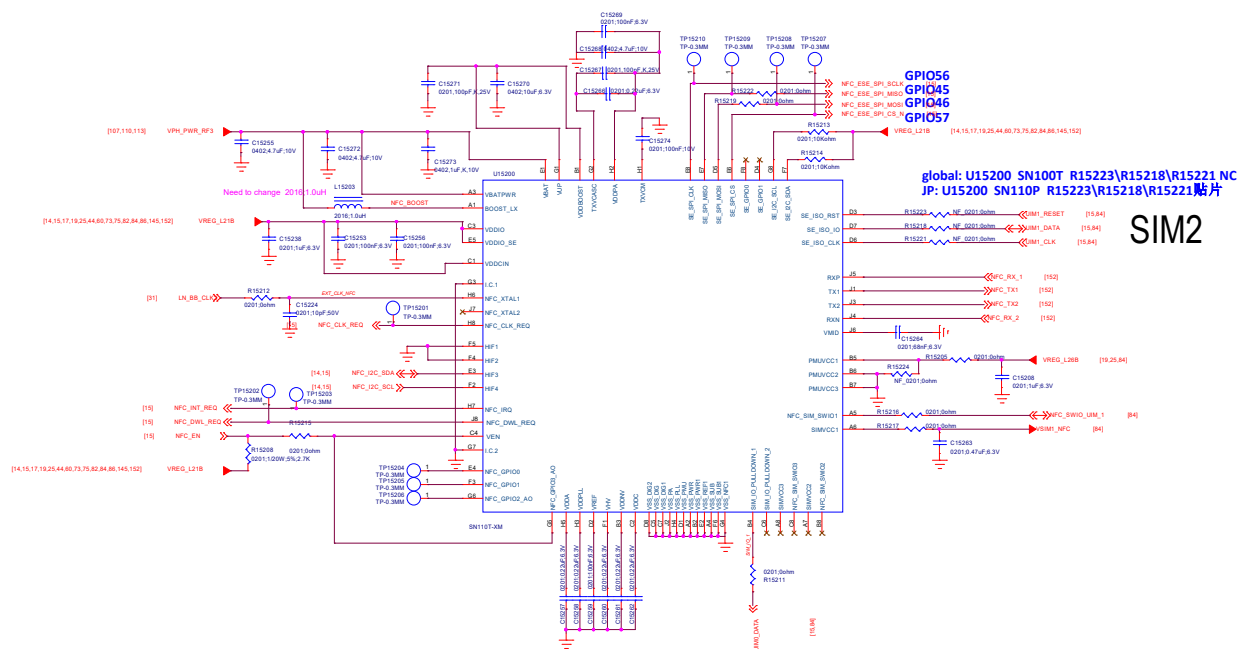
150.GPS



Title		REV: V30
DOCUMENT NO.: Design Name = 88829 - I_12M14_2023041		8885 D
DEPARTMENT:	DEPARTMENT = RF_32as	DESIGNER: DESKNER - XA0506
		
Date:	Page Modify Date = Monday, April 17, 2023	Sheet 250 of 256

152.NFC_SN100T

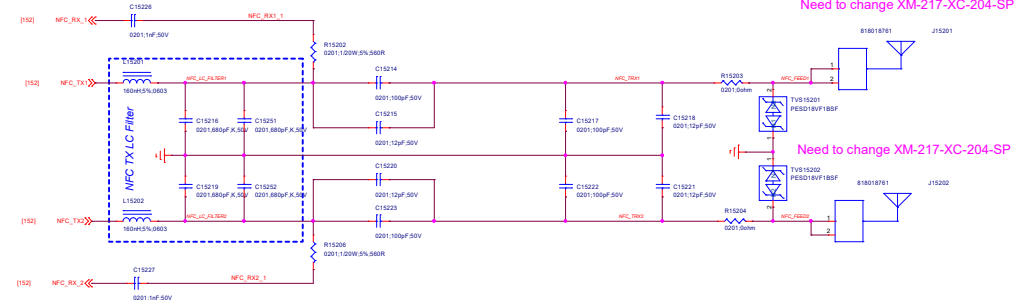
换成 SN100T



SIM2

Need to change XM-217-XC-204-SP

Need to change XM-217-XC-204-SP



Title		REV: V30
DOCUMENT NO.: Design Name = 88829_1_12M14_2023041		8885 D
DEPARTMENT:	DEPARTMENT - RF_30m	DESIGNER: DESIGNER - XAC
		
Date:	Drawn/Modify Date = Monday, April 17, 2023	Sheet 152 of 156